Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 1, v.3.4 (October 28, 2003)

1. From Atoms to Planets: Scales of Size

Imagine a large sheet of millimeter graph paper, of size 1 m x 1 m, glued to the wall. Stare at it and, if you have reasonably good vision, you will be able to see, at one time, a million little squares. One thousand in row No.1, one thousand in the next row – that's two thousand, so far. Then, one thousand more little squares in each of the other rows, until we reach the last thousand squares in row No.1000. A thousand thousands = one million little squares!

This may well be the largest single number that it is possible to see at a single glance. Why? Because although a larger sheet of graph paper - say, 3 m x 3 m - would contain more squares (How many?), one would have to step back in order to see the entire sheet in focus but, in so doing, one would probably not be able to resolve the individual 1 mm squares.

But let's go back to our 1 m x 1 m sheet, and cut it into 1000 strips, each 1 mm wide and 1 m long. If we place these 1000 strips end to end we will have a strip 1 km in length.

We can thus visualize the relationship between 1 mm and 1 km: There are 1,000,000 mm in 1 km, therefore, 1 km is 1,000,000 times larger than 1 mm. It is convenient to write the number 1,000,000 as 10^6, where the superscript "6" indicates the number of times that 10 has been multiplied by itself. (Alternatively, it represents the number of zeros after the 1). We can also see that 1 mm is 1,000,000 times *smaller* than 1 km, or, 1/1000,000 times as large as 1 km. We may write the fraction 1/1,000,000 as $1/10^6 = 10^{-6}$, where the superscript "-6" indicates how many zeros there are in the denominator.

If we wanted to measure the world in millimeters we would have to refer to 1 m as 10^3 mm and 1 km would be 10^6 mm. This is clearly inconvenient and it is the reason why different scales of length have been invented (i.e., purely out of convenience) for measuring different scales of distance. We use kilometers to measure the distance between cities, meters to measure the size of buildings and millimeters to measure small, visible, objects in a machine shop. Of course, astronomers and atomic scientists naturally find other scales of length more convenient for their work. Let's see what kinds of distance they need to measure.

Since we now feel perfectly comfortable with the concept of one million or 10^6 (i.e., 1 km is simply a million times larger than 1 mm, or 1 km = 10^6 mm), let us apply it to the kilometer, used as a starting point and increase this unit by a factor of one million.

Q. What use do you think might be made of a length scale of 10^6 km, or 1 *Giga*meter = 1 Gm?

A. The mean distance between the earth and the moon is 0.3844 Gm. So the Gm could be a convenient scale for measuring the distance between planets and their satellites.

We could also increase the Gigameter by another factor of a million. This would give us one *Peta*meter, or 1 Pm = 10^6 Gm = 10^{12} km = 10^{18} mm. The Petameter is a very large distance indeed. It is approximately 1/10 of a *light-year*, i.e. the distance that a beam of light would travel in approximately 40 days.

Now let us go in the opposite direction.

Q. What is one million times smaller than 1 mm, or 10^{-6} mm?

A. If we could see such small distances with our eyes we would be able to see the individual atoms laid out in regular arrays within all kinds of crystals. For this length scale, also called 1 *nano*meter or 1 nm, typifies the distances between atoms. For example, in table salt, the atoms of Na and Cl are arranged in a series of perfect interlocking cubes - just like the crystals themselves - the distance between adjacent atoms of the same kind being 0.563 nm. Today, so-called ‘‘nano-technology” is a buzzword. It refers to engineering devices at the scale of the interatomic distances in crystals.

But we needn’t stop there. If we decrease the nanometer by another factor of a million we arrive at the *femto*meter. 1 fm = 10^{-6} nm = 10^{-12} mm. This is approximately the size of the protons and neutrons that reside in the nuclei of atoms.

Summary

We see that "one million", which is the ratio between the two familiar length scales of 1 mm and 1 km, is also the ratio of 1 km to the start of astronomical distances and, conversely, the ratio of inter-atomic distances to 1 mm. The good news is that, in this course, we shall not require distance scales smaller than nanometers, or larger than Gigameters!

2. The role of atoms in a geography course?

The purpose of this course is to introduce you to the physics of the environment. Now, consider an automobile. Anyone can learn to drive and enjoy the benefits of an automobile without having any idea how one works. In fact, this is precisely what most of us do! On the other hand, a garage mechanic does understand how an automobile works because he understands the function of the parts out of which it is built. Well, we have a similar situation in geography. We can all enjoy mountains, lakes, fresh air, etc., without having any idea "how they work", but by understanding something about the atoms out of which they are made we shall come to appreciate the physics of the environment in which we live.

In each lecture I shall, therefore, endeavor to pick at least one geographical phenomenon and relate it to what the underlying atoms are doing.

Now usually this would require a considerable mastery of mathematics, which few of you are likely to have. So I shall simplify matters by developing any necessary mathematical techniques as we go along.

3. Pythagoras’ theorem and square roots

You are all familiar with the famous theorem about right-angled triangles: "The square on the hypotenuse equals the sum of the squares on the other two sides". Or, in symbols:

$$A^2 + B^2 = C^2 \qquad (\mathbf{1.1})$$

For example, if A = 3 and B = 4, then $A^2 + B^2 = 9 + 16 = 25$ and hence, C = 5. Such a triangle obeys Pythagoras' theorem and is consequently a right-angle triangle.

But the sides do not have to be expressible as simple integers in order for them to obey the theorem. For example, if A = 2 and B = 3 and the angle between them is 90º then, from Pythagoras' theorem, $C^2 = 4 + 9 = 13$ and consequently $C = \sqrt{13}$.

In order to convert this to a usable number you must use your calculators. The result is then 3.6 or 3.605 or 3.60555, etc., depending upon how accurately you need to know the answer.

But the rules of mathematics sometimes enable us to evaluate even quite complicated numbers rather simply. In particular, Pythagoras’ theorem presents us with a method for evaluating

certain square roots without the need for a calculator – specifically, ones in which B is small compared with A. For such situations the theorem becomes:

$$\sqrt{(A^2 + B^2)} \approx A + B^2/2A \qquad (\mathbf{1.2})$$

where the wiggly symbol means, "approximately equal to".

This expression is relatively easy to evaluate without a calculator, but it only works when B is "small". How small is small? We shall need to examine a few cases in order to find out.

First case: B = 0.5, A = 1.
Then C = $\sqrt{1.25}$
Our calculator tells us that C = 1.118, whereas eq. (**1.2**) gives 1.125 (but the error is < 1%)

Second case: B = 0.1, A = 1.
Then C = $\sqrt{1.01}$
Our calculator tells us that C = 1.0049875, whereas eq. (**1.2**) gives 1.005

Third case: B = 0.05, A = 1.
Then C = $\sqrt{1.0025}$
Our calculator tells us that C = 1.0012492, whereas eq. (**1.2**) gives 1.00125

Fourth case: B = 0.01, A = 1.
Then C = $\sqrt{1.0001}$
Our calculator tells us that C = 1.0000499, whereas eq. (**1.2**) gives 1.00005.

I have not gone any further because, at this stage, the approximate formula is now as accurate as the 8-digit calculator I am using!

I have drawn your attention to this particular mathematical trick because it turns out to be enormously useful in many areas of physics – including one we shall now examine.

4. Pythagoras' Theorem and Some Basic Physics

From time to time I will have to "throw" a physics formula at you, since this is a physics course and since there will not be enough time for me to derive the formula in a manner that would enable you to understand where it came from. However, whenever I do this, I shall try to explain the formula as well as I can.

One of the most remarkable physics formulae to have emerged in the 20th century has a mathematical form that is identical to Pythagoras' theorem:

$$m^2 + p^2 = E^2 \qquad (\mathbf{1.3})$$

This formula applies to any freely-moving particle, i.e. one that is not moving under the influence of any external forces. This formula relates the *energy* E of a particle to its *mass* m and its *momentum* p. (For the time being, let us not worry about what these italicized words mean. I'll get back to this later). Now the formula need not scare us because, mathematically, it is no more difficult than Pythagoras' theorem. But it contains a wealth of information about the physical world!

Basically, the formula in equation (**1.3**) makes 3 strong statements:

(a) If a particle is not moving it has no momentum (i.e. p = 0), then:

$$E^2 = m^2 \qquad (\mathbf{1.4})$$

(b) If a particle has no mass - amazingly, there are such particles (i.e. m = 0) - then:

$$E^2 = p^2 \qquad (\mathbf{1.5})$$

(c) A particle can never have zero energy (otherwise $m^2 = - p^2$, which is a mathematical impossibility, since the square of any real number can not be negative).

Now what happens if we have a slowly moving particle? I.e. if p is very small and, hence, very much less than m. Then, from formula (**1.2**) we expect:

$$E = m + p^2/2m \qquad (\mathbf{1.6})$$

Eq. (**1.6**) tells us that for a slowly moving particle the energy equals the sum of a term that depends on its momentum (we call this term "kinetic energy") and another term which is apparently simply the mass of the particle (high energy physicists call this term the "rest mass energy"). We shall look more closely at the meaning of energy, momentum and mass in the next lecture.

Problem set No.1 (numbers, dimensions and units)

1. A particular kind of solar collector is rectangular with dimensions 1 m x 2 m. What is the maximum number of solar collectors that could be laid out horizontally in a square field of dimensions 1 km x 1 km, without the collectors shading one another?

2. Suppose that a crystal has the shape of a cube with dimensions 1 cm x 1 cm x 1 cm. If the individual atoms from which the crystal is made are represented by hard spheres of diameter 0.1 nm, and they are packed tightly together so that they touch one another, how many atoms would the crystal contain?

3. Use Pythagoras' theorem in order to estimate the value of $\sqrt{110}$ *without* using a calculator. What is the percentage error in your estimation? You may use a calculator for the second part.

4. In British units, 1 mile = 63,360 inches. 1 inch is defined as 25.4 mm exactly. There are precisely 3600 seconds in one hour. How large is the speed of 100 miles per hour (mph) in meters per second (ms^{-1})?

5. The so-called "fine structure constant" is a dimensionless number defined as:
$\alpha = (e^2/4\pi\varepsilon_0)/(h/2\pi)/c$, where

$e = 1.602 \times 10^{-19}$ C is the amount of electric charge on an electron,
$\varepsilon_0 = 8.854 \times 10^{-12}$ C^2 kg^{-1} m^{-2} s^2 is a certain electrical property of the vacuum,
$h = 6.626 \times 10^{-34}$ kg m s^{-1} is a constant of nature known as the "Planck constant"_
$c = 2.998 \times 10^8$ m s^{-1} is the speed of light in a vacuum,
$\pi = 3.142$ is the ratio of the circumference to the diameter of a circle.

Calculate the value of α, and check that your result is dimensionless.

Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 2, v. 3.3 (November 4, 2003)

1. Momentum

In lecture 1 we encountered a number of terms that are probably semi-familiar to some of you but, perhaps, not yet fully understood. Others among you may have merely heard the words "mass", "momentum" and "energy", but never encountered them in a scientific context. One of the aspects of science that often confuses non-scientists is that it tends to take commonly used words and give them an extremely specialized definition. This is the case for these three words, so let me now begin to try and clarify their meaning as used in the context of physics.

Last week we found that, by applying Pythagoras' theorem to one of the equation's that Einstein discovered, a slowly-moving, free particle has two components to its total energy. One component is denoted by the symbol m. This component is always present, no matter how quickly or slowly the particle may be moving: even when it is completely at rest. For this reason it is known as the *rest-mass energy*. However, if the particle is moving slowly, it has an additional energy component due to its motion. This component is called the *kinetic energy* and is equal to the quantity $K = p^2/2m$, where p is the so-called *momentum* of the particle.

Now you may be more familiar with another formula for kinetic energy:

$$K = mv^2/2 \qquad (\mathbf{2.1})$$

where v is the *velocity* of the particle. Velocity is a more familiar everyday quantity because it is something one can actually see. Velocity is the *rate at which the position of a particle is changing*. One moment it is here; a second later it is 5 m away; another second later it is 10 m away; still another second later it is 15 m away: it's position is therefore changing at the rate of 5 m each second - we say that its velocity is 5 m/sec or 5 m sec^{-1}.

Momentum, on the other hand, is less familiar as an everyday phenomenon, so it will require some further discussion. But before we start, it is important to realize that since kinetic energy can be described either in terms of the velocity v, as in formula (**2.1**) or, alternatively, in terms of the momentum p, via the expression:

$$K = p^2/2m \qquad (\mathbf{2.2})$$

Then, it must be that:

$$p = mv \qquad (\mathbf{2.3})$$

We may take equation (**2.3**) as a working definition of momentum: *momentum equals mass times velocity*. But it is rather a "dry" formula so let us now try and put some life into it.

If you fire a gun you notice that the gun jumps backwards as the bullet flies forwards. Now you could make a movie of this phenomenon in slow motion, and use the film frames to calculate the velocity of the bullet in the forward direction and the velocity of the gun in the backward direction - at the moment it fires. You would then find something very interesting: the forward velocity of the bullet multiplied by the mass of the bullet would equal the backward velocity of the gun multiplied by the mass of the gun. In other words, *the forward momentum of the bullet equals the backward momentum of the gun*.

Experiments, performed on all kinds of objects, with ever increasing accuracy, show that this is always true: If a stationary object breaks up into a number of parts then the sum of all their momenta - when due account is taken of their various directions - will be zero, i.e. the

momentum of the object before it broke up. On the other hand, if a moving object breaks up while in motion, the sum of the momenta of its parts will equal the momentum of the object before it broke up. This law of nature is called the *Law of conservation of momentum*. It is the reason that guns are heavy and bullets are light: we want the bullet to move rapidly and the gun to move slowly! If we want to use a heavy bullet (say for piercing armor plate) then we must use an even heavier gun.

The Law of conservation of momentum is a universal law that applies to <u>everything</u>: exploding stars, exploding bombs, exploding atoms and even exploding sub-atomic particles.

Two out of the many known sub-atomic particles are the *rho-meson* and the *pi-meson*. A rho-meson does not last very long after it is created - about 10^{-24} sec in all. It then breaks up into two pi-mesons. If the break-up occurs when the rho-meson is stationary then the resulting two pi-mesons fly off in exactly opposite directions with exactly equal, but opposite, velocities. This is because momentum must be conserved. The reaction may be written like a chemical reaction:

$$\rho^o \rightarrow \pi^+ + \pi^- \qquad (\mathbf{2.4})$$

where the superscripts in equation (**2.4**) indicate the electric charge that each pi-meson carries.

Now there are actually three charge varieties of pi-meson in nature: the third is electrically neutral and also unstable but it has an extremely interesting (for our purposes) way of breaking up. In contrast to the rho-meson, the π^o meson "lives" an incredibly long time - about 10^{-16} sec! Like the rho-meson, the pi-meson also breaks up into two particles, but these are *photons*. The reaction may be written:

$$\pi^o \rightarrow \gamma + \gamma \qquad (\mathbf{2.5})$$

where, once again, the two photons fly off in opposite directions with equal velocities because of the law of conservation of momentum.

But in reaction (**2.5**) we encounter something almost unique: The photon is a particle that has momentum but no mass! How is that possible if $p = mv$? You are correct: it is a mathematical impossibility unless the photons move at infinite velocity - which they do not. Photons move at the *velocity of light*, (denoted by the symbol c, and equal to the universal constant: 299,792,458 m sec^{-1}) because they *are* light. We should therefore expect them to have zero momentum if they have zero mass.

The answer to this riddle is that $p = mv$ is only strictly true for *slowly-moving* particles - you will remember that we derived eq. (**2.2**) for a slowly-moving particle, using our small-angle approximation to Pythagoras' theorem. For particles that move with velocities that are an appreciable fraction of the velocity of light, that approximation is no longer very accurate. Therefore, it should not be surprising that the formula $p = mv$ would need to be modified. However, in geographical situations we shall never encounter a particle with mass that moves fast enough for us to have to worry about what the accurate formula is for momentum, so I shall not even write it down.

For our purposes the important things to know about photons are:

(**1**) They behave like any other sub-atomic particles.
(**2**) They happen not to have any mass (just as they happen not to have any electric charge).
(**3**) They always travel at the speed of light. (This turns out to be a consequence of their not having any mass, but that is the *theory of relativity*, which we shall not need to learn for purposes of the present course).
(**4**) They are the constituent particles out of which light and all other forms of electromagnetic radiation (e.g. gamma rays, X-rays, UV radiation, IR radiation, Radar, Microwaves, Radio

waves, etc.) are made.

(**5**) In addition to behaving like particles (e.g. they can move freely through empty space), they also have wavelike properties. (It turns out that so too do all of the other sub-atomic particles that do have mass. But an understanding of this fact – *quantum theory* - also lies outside the requirements of the present course).

We shall return to properties (**4**) and (**5**) in later lectures because the light and heat energy that reach us from the sun arrive in the form of streams of photons.

2. Mass

We have talked a lot about "mass" without saying what it is. All we know so far is that it is a property of particles, small and large (although some subatomic particles, like photons, evidently do not have any mass). But mass is easy to measure because it is also a property of large objects. It is a measure of how much of a particular substance we have. If we place two identical gold coins on the pans of a scale, they remain balanced because they both have equal amounts of mass. If we replace one of the coins by a smaller gold coin, that scale pan goes down because the smaller coin has less mass than the large coin. One may define a convenient unit for measuring the mass of coins, e.g. the gram. One can then use this unit, for example, to quantify how much mass there should be in a particular kind of gold coin (e.g. 10 gm), and, thereby, to help detect forgeries.

For more massive objects, the gram is inconveniently small, so, in like manner for measuring distances, we can define the kilogram, the megagram (better known as a tonne), the gigagram, etc. … Similarly, for assessing the mass of extremely light objects we may use, milligrams, micrograms, nanograms, etc.

For most purposes in this course (i.e. unless stated to the contrary) we shall employ the kilogram (kg) as our unit of mass, the meter (m) as our unit of length, and the second (s) as our unit of time. This combination is known as "*S.I. units*". All other mechanical quantities, such as force, energy, power, etc. will then be defined in terms of appropriate combinations of these three basic units

3. Weight

In daily life, we also use an expression "weight" which is easy to confuse with "mass" because the same units are commonly used for both. But, as we shall see in a moment, the two concepts are very different.

Recall that we measure mass by placing the object we wish to assess on one pan of a pair of scales, and placing a succession of standard "weights" on the other pan until the two pans are found to balance. In this way we discovered above that a particular kind of gold coin has a mass of 10 gm. But we could also use a different kind of measuring instrument, based on the concept of stretching a spring. This, so-called, "spring-balance" has a graduated scale attached to it, with grams marked out. Thus, if we place the same gold coin on the pan of a spring-balance, we shall find that the spring extends until the pointer ends up at the mark 10 gm.

Now suppose that we were to repeat these two measurements on the Moon. We would find the (perhaps) surprising result that, whereas the 2-pan pair of scales would still indicate the mass of the gold coin as being 10 gm, the spring-balance would indicate only 1.65 gm!

Why would we receive such a surprising result? The reason is that a spring-balance actually measures not the mass of the gold coin but, rather, the *force* that *gravity* exerts on it. Force and gravity are two more of those common words that are used by physicists in very specific ways. We shall define them more precisely in later lectures. As we shall also learn, because the force due to gravity is smaller on the Moon than it is on Earth, the spring balance would indicate a smaller reading than the 2-pan scales. If this is so, why would there be no change in the 2-pan

scales? Because, the lower force of lunar gravity would act in precisely the same way on the masses in both pans. Therefore, they would remain balanced.

We may say, therefore, that a 2-pan pair of scales measures true mass, whereas a spring-balance measures the force of gravity on that mass. This force we call *weight*. It is traditional to use the *same units* for mass and weight, which is why there may be some confusion. This tradition has grown up because in the past, nobody had ever left the Earth. But one day, when inter-planetary travel becomes common, it will be necessary to make a careful distinction between mass and weight – particularly, if you are the buyer!

Strictly speaking, *weight equals mass times the acceleration due to gravity*. In symbols we may write this as:

$$W = m\,g \qquad (\mathbf{2.6})$$

where g is the acceleration caused by gravity (which we shall define and discuss in a later lecture). The number g is approximately constant for all parts of the earth's surface, and equal to 9.8 m s^{-2}. Therefore, in a system of units in which a certain object has a mass of 10 kg, its weight should really be expressed as 98 N, where the Newton, N, is a unit of force in S.I. units, equal to 1 kg m s^{-2}. However, because g is essentially a constant everywhere on earth, it is more convenient to treat the mass of an object as its weight. This is equivalent to assuming that g = 1, but may sometimes cause confusion if you forget that mass is the intrinsic property of a body and that weight is really the force that gravity exerts on it.

4. Mass and Energy

If we return to eq (**1.6**) from the first lecture:

$$E = m + p^2/2m \qquad (\mathbf{2.7})$$

and check the dimensions and units, we see that the kinetic energy term has dimensions $m\ l^2\ t^{-2}$, i.e. mass times velocity-squared, which in SI units, would be kg m^2 s^{-2}. This unit of energy is called the *Joule*. However, one term in eq. (**2.7**) apparently has the wrong units. Mass again! This time we are not guilty of dropping a g because that constant would give mass the dimensions of a force. In order for it to have the dimensions of an energy it must be multiplied by a constant with the dimensions of a velocity squared. Although this will seem highly arbitrary in a geography course, the correct constant happens to be the square of the velocity of light in free space. This number has been measured with great accuracy and is now defined as being precisely the number given above for c. For our purposes c = 3 x 10^8 m s^{-1} is an excellent approximation. Thus, eq.(**2.7**) should actually read:

$$E = mc^2 + p^2/2m \qquad (\mathbf{2.8})$$

This is precisely the same as eq. (**2.7**) if we are using units in which c =1. Similarly, eq.(**1.3**) from lecture 1 should really have been written as:

$$E^2 = m^2c^4 + p^2c^2 \qquad (\mathbf{2.9})$$

Eq.(**2.9**) is correct for all possible particle velocities, fast as well as slow, and the dimensions are now such that m is indeed a mass, p is a momentum and E is an energy. However, physicists who actually deal with "high energy" particles every day prefer to use "energy units" in which c = 1. Similarly, we, who deal with masses of kilograms every day, prefer to use units of weight in which g = 1. It is simply a matter of convenience.

Returning now to eq. (**2.8**), having now unified the dimensions of all the terms and settled upon units of Joules for energy, it is instructive to calculate the relative amounts of kinetic energy and

rest-mass energy in, say, a 10 g bullet moving at 1000 m sec^{-1}. First, the kinetic energy of the bullet is:

$$mv^2/2 = 0.01 \text{ kg} \times (1000 \text{ m s}^{-1})^2 / 2$$

$$= 5{,}000 \text{ Joules} \qquad (\mathbf{2.10})$$

On the other hand the non-kinetic, rest-mass term takes the value:

$$mc^2 = 0.01 \text{ kg} \times (300{,}000{,}000)^2$$

$$= 9 \times 10^{14} \text{ Joules} = 900{,}000 \text{ GJ} \qquad (\mathbf{2.11})$$

The quantity in (**2.11**) is a huge amount of energy. If it were converted into electrical energy it would equal 250,000,000 kWh which is approximately the amount of electricity that could be generated by a 30 MW power station operating non-stop for an entire year – i.e., enough for the entire city of Eilat. And all of this energy resides inside the mass of a 10 g bullet, independent of what speed it is moving!

Could the mass of a bullet actually be converted into a useful form of energy? The answer is: Yes. We call it *nuclear* energy, and we already saw an elementary example of the conversion of rest-mass energy to radiant energy in the reaction given in eq (**2.5**) where a "massive" pi-meson decays spontaneously into pure radiation - in this case, two photons.

Nuclear processes are responsible for the energy emitted by the sun and also for geothermal energy generated within the earth's interior. They are examples of mass being converted into radiation. We shall return to such matters in later lectures.

But why are we unaware of this massive amount of energy in everyday life? The answer is partly that it is locked up so strongly inside each piece of matter that it cannot be released unless we make an extraordinary amount of effort to get at it (i.e. unless we cause a thermo-nuclear reaction to occur). Secondly, in our everyday experience of energy we really only measure changes in energy. The bullet contains all of this rest-mass energy whether it is stationary in the gun or whether it is fired, but it is the kinetic energy of the bullet that is the quantity of practical use: a stationary bullet can not do any damage, but a rapidly moving bullet is another story.

In later lectures we shall encounter other forms of energy that are useful on a daily basis (e.g. potential energy, thermal energy, chemical energy, etc.) but in all cases we shall see that it is only *differences in energy* that are of interest: we shall rarely, if ever, have to calculate an absolute quantity of energy.

For this reason, after the present lecture, we shall simply forget about the existence of rest-mass energy. It will be of no further interest to us (except at examination time!).

Problem set No. 2 (momentum and energy)

1. Suppose that instead of taking mass (m), length (l) and time (t) as our fundamental dimensions, we were to adopt new ones based on : velocity (v), force (f) and energy (e). What would be the dimensions of mass length and time in terms of new dimensions?

2. A 1 km object has a velocity of 100 m s^{-1}, and a 1 tonne object has a velocity of 0.1 m s^{-1}. Which object has:

   (a) the larger momentum?
   (b) the larger kinetic energy?
   (c) the larger total energy?

3. A 2 kg block of wood rests upon a frictionless horizontal surface. A 10 gm bullet, moving horizontally at 1000 m s^{-2}, embeds itself in the wooden block. With what velocity does the block begin to move after being struck by the bullet?

4. An astronaut has a mass of 90 kg when measured on earth. If the acceleration due to gravity on earth takes the value $g_E = 9.81$ m s^{-2}, and its value on the moon is $g_M = 1.62$ m s^{-2},

   (a) What is the weight of the astronaut on earth?
   (b) What is his mass on the moon?
   (c) What is his weight on the moon?

5. You are an astronaut, having a race with another astronaut. Your two space ships are travelling parallel to each other in space and both have the same speed of 1000 kph relative to the earth. Each of the space ships has a total mass (i.e. including crew and all on-board equipment) of 5 tonnes.

   You look sideways out of your window and see that the astronaut in the other space ship has a nasty smile on his face and is holding a 1 tonne object in his hand. He opens his window, slowly moves the heavy object outside, holds it between his ship and yours, and releases it!

   (a) Does the object appear to move forwards, backwards, or stand still ?
   (b) Why ?
   (c) Does the other space ship appear to move faster, slower or remain at the same speed?
   (d) Why?

   This gives you an idea! You now take hold of a 1 tonne object, open your window, place the object outside, but project it backwards at a speed of 10 m s^{-1} relative to your space ship.

   (e) Does the other space ship appear to move backwards, forwards, or remain stationary relative to your ship?
   (f) Why?
   (g) How does the 1 tonne weight that was ejected by the other astronaut now appear to move?
   (h) Why?
   (i) Calculate the values of the various speeds you have concluded.

Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 3, v. 1.7 (November 18, 2003)

1. Force and Newton's Laws of Motion

Let us return, yet again, to eq (**2.9**) from the last lecture, (which is the same as eq (**1.6**) from the first lecture but with all terms having the correct dimensions):

$$E = mc^2 + p^2/2m \qquad (\mathbf{3.1})$$

It will be slightly more helpful, in the following discussion, to write the kinetic energy term using velocity instead of momentum:

$$E = mc^2 + mv^2/2 \qquad (\mathbf{3.2})$$

The kinetic energy term is not an intrinsic property of the particle since it depends upon how *we* measure its velocity. This may seem strange, however, if we run along side the particle at the same speed then we shall observe that its velocity seems to be zero. Hence, it will not have any kinetic energy. It will still have its built-in "rest-mass energy" since that part does not depend upon its motion.

Now all the time that the particle remains in uniform motion, i.e. v = constant, no real meaning can be attached to its velocity. If we were observing it in free space we could not tell whether the particle or we were moving. You may have noticed similar effects on a boat at sea, or even while sitting in a bus and waiting for it to move. As the bus next to us moves forward we get that momentary strange feeling that we are moving backward? This is an example of the **Principle of Relativity** which states: *There is no experiment you can perform that will enable you to know the absolute velocity of a uniformly moving object.*

Similarly, *a uniformly moving object will continue that way forever unless acted on by some external force that changes its velocity* (i.e. speeds it up, slows it down or changes its direction). This is a statement of what is known as **Newton's First Law of Motion.**

But what do we mean by "force"? In order to understand this concept, in a quantitative manner; let us return to the definition of velocity as the *rate of change of* position. It is convenient to visualize this graphically.

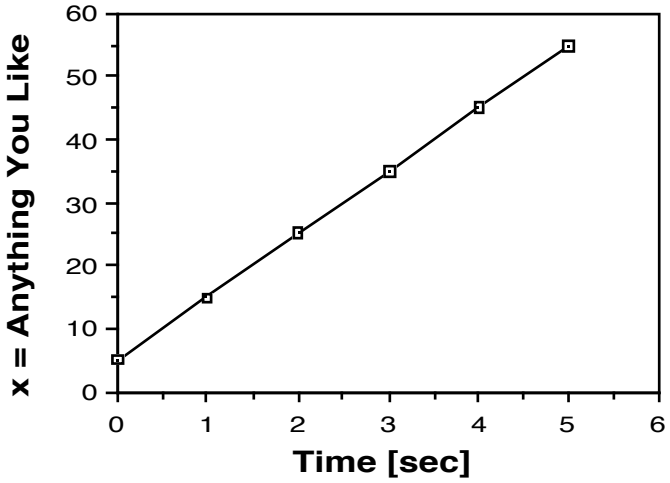


Figure 1: Graph of "anything" vs. time. If "anything" represents *position*, the slope of the graph gives the object's velocity. If "anything" represents *velocity*, the slope of the graph gives the object's acceleration.

Let x indicate the position of a particle, and t the time. If the particle moves at constant velocity then equal changes in x will occur during equal intervals of time. In the particular example shown on the graph in **Fig. 1** the particle is at x = 5 m (or cm or km or anything you like) at t = 0. After 1 sec it has moved to x = 15 m. After another 1 sec it is at x = 25 m, etc., etc. We see that its velocity is 10 m s^{-1}, i.e. its position is changing uniformly at the rate of 10 m each second. This uniform velocity is represented on the graph by a straight line of a definite slope angle. If the velocity of the particle were larger - e.g., 20 m s^{-1} - the slope of the graph would be steeper. (Plot it and see!). On the other hand, for a smaller velocity the slope would be gentler. In fact, the slope would be zero if the particle were stationary. Why? Because at times t = 0, 1, 2, ..., etc., it would always have the same value of x.

What would happen to the graph if the particle's velocity were not constant? If the particle accelerates its velocity increases. This means that the slope of the graph would increase and it would no longer be a straight line. But let us stay with straight-line graphs for the time being.

I labeled the vertical axis of the graph in **Fig. 1** "x = Anything You Like" because x does not have to represent the position of the particle. We could equally well use such a graph to represent the way velocity changes with time. Thus, in the graph in **Fig. 1**, where x now represents velocity, at time t = 0 the velocity is 5 m s^{-1} (or any other units you may prefer), at t = 1 sec the velocity has increased to 15 m s^{-1}, at t = 2 sec the velocity is now 25 m s^{-1}, etc., etc. We see that the velocity is increasing at a steady rate of 10 m s^{-1} each second - or, expressed mathematically, at the rate of 10 m s^{-2}. In this case, the slope of **Fig. 1** represents the *acceleration* of the particle.

These straight-line graphs, when something is plotted as a function of time, are very helpful for visualizing quite abstract concepts. Velocity is an everyday phenomenon so it is reassuring to find that a constant velocity is represented by a straight line on a position-time graph. Acceleration is a slightly more abstract concept. We can "feel" it but the velocity-time graph gives us a quantitative measure for acceleration, and if the latter is constant then the linearity of the plot enables us to see, at a glance, precisely what the velocity is at every instant of time.

If the acceleration were zero, then the velocity-time graph would be a horizontal straight line. I.e. at times t = 0, t = 1, t = 2, ..., etc. the velocity would always have the same value.

Now, if we multiply the particle's velocity by its mass we obtain its momentum. So the same horizontal graph of momentum plotted against time would tell us that the particle was moving with constant momentum.

But what if the momentum graph were to have a positive slope? This would tell us that the momentum was increasing at a constant rate. This, in turn, would indicate that the particle was accelerating. What would be the cause of such an increase in momentum or acceleration? The answer is that some external force must be acting on the object. **Newton's Second Law of Motion** states that *force is the rate of change of momentum*. This law actually provides a quantitative definition of the concept of force. Alternatively, since momentum = mass x velocity, it is also correct to define force as the mass of a body multiplied by its acceleration:

$$F = ma \qquad (\mathbf{3.3})$$

If mass is measured in kg units and acceleration is measured in units of m s^{-2}, then force is measured in "newtons", denoted by the symbol N.

Eq. (3.3) is only true if the object has a constant mass. This assumption would not be true for a rocket that is burning fuel as it accelerates and therefore has a decreasing mass. In that kind of situation one must calculate the rate of change of momentum where both the velocity and mass are changing. But, in this course we shall not encounter such complicated problems.

Now, unlike velocity, acceleration is *not* relative. If you are in free space and looking at an accelerating object its velocity will appear to be increasing relative to you. But how do you know that it is not you who are accelerating? Simple - take a bucket of water with you. If the water level remains smooth you are not accelerating. Did you ever try drinking some hot tea in a bus, at precisely the moment the driver decides to move out of the bus station? You are the one who accelerates: not the girl sitting in the other bus and laughing!

Newton's Third Law of Motion is the observation that *active and reactive forces are equal and opposite*. What does this mean? Simply, that if you stand on the ground the ground pushes up on you with a "reactive force" that is of exactly the same magnitude as the force with which you push down on the ground. If this were not the case then you would either move upward or downward. An elevator floor, for example, is capable of pushing up on you with a greater force than you push down on it. This enables it to raise you to whichever floor you would like to reach. Other buttons enable you to exert a greater force on the elevator than it exerts on you and, hence, to go down.

Newton's three laws of motion are very general, in that they apply in all situations one encounters in everyday life. They constitute a complete formulation of the science of "mechanics". There are also several other alternative formulations – that physics students have to learn – but because Newton's laws are intuitively rather clear, they are pedagogically the most useful.

2. Gravity

Newton's three famous laws of motion are also very general in that they apply no matter what kind of forces are at work (with certain very specific kinds of exception that are outside the scope of this course since they are of no relevance to geography). But Newton also discovered a very specific force law for the case of gravitational forces. By observing the motions of the sun, moon and planets he realized that their varied and complicated motions could all be explained if the force with which any two bodies attract one another has two simple properties:

(1) It is proportional to the product of their masses.

I.e. double the mass of one of them (either one - it does not matter which) and the gravitational force doubles; double the mass of one and triple the mass of the other then the gravitational force becomes six-times as great; etc., etc.

(2) It is inversely proportional to the square of the distance between them.

I.e. double the distance between them and the gravitational force decreases to a quarter of its original value; triple the distance between them and the gravitational force decreases by a factor of nine; etc., etc.

Newton's Law of Gravity may be written:

$$F = (G\, m_1\, m_2) / r^2 \qquad (\mathbf{3.4})$$

where, m_1 and m_2 are the masses of the two bodies, r is the distance between them, and G is a constant that must be fixed according to the units one is using. If the masses are measured in kg and the distance between them is measured in m, then Newton's so-called **gravitational constant** takes the value $G = 6.67 \times 10^{-11}\ m^3\ kg^{-1}\ s^{-2}$.

For terrestrial purposes, i.e. the gravitational attraction of the Earth to objects on its surface (like you and me), it is more convenient to make use of another constant, g **the acceleration due to gravity**. In order to see the relationship between G and g it is convenient to consider the gravitational attraction between the Earth and an object on its surface. We have:

$$F = G\,M\,m / R^2 \qquad (\mathbf{3.5})$$

where M is the mass of the Earth (5.97×10^{24} kg), m is the mass of the object on its surface, and R is the mean radius of the Earth (6370 km). On the other hand this force of attraction is simply equal to the weight of the object, which, in the last lecture, we defined as:

$$F = m\,g \qquad (\mathbf{3.6})$$

By equating these two expressions we see that the weight cancels out, and:

$$g = GM/R^2 = 9.81 \text{ m s}^{-2} \qquad (\mathbf{3.7})$$

The constant g is equal to the acceleration with which a particle falls if you release it near the Earth's surface. Acceleration is, of course, *the rate of increase with time* of velocity. If the object is released at rest, then after 1 sec. its velocity will be 9.81 m s^{-1}. After 2 sec. its velocity will have increased to 19.62 ms^{-1}. After 3 sec. its velocity will have increased to 29.43 ms^{-1}, etc.

In a later lecture we shall see that because the Earth is rotating and not uniformly spherical in shape, the precise value of g will vary very slightly from place to place. Therefore, for purposes of calculations in this course, the value 9.8 m s^{-2} will be of adequate accuracy.

Problem set No.3: **(forces, etc.)**

1. In **Figure 1**, suppose that **"Anything You Like"** represents the velocity of an automobile in units of km per hour. What is its acceleration in m s^{-2}.

2. If the mass of the automobile in the previous question is 750 kg, what force is its engine exerting on it in order to produce such an acceleration?

3. What is the "weight" of the above-mentioned automobile, in units of: (a) kg; (b) N ?

4. The planet Mars has a mass which is 10.7% that of the Earth, and a radius that is 53.2% that of the Earth. Calculate the value of g, the acceleration due to gravity, on the surface of Mars.

5. If a person weighs 90 kg on Earth, how much would he weigh on Mars?

1. Circular motion and centripetal force

Newton's First Law told us that an object continues to move in a straight line unless acted upon by an external force. A special case of an external force arises in the case of an object that moves, at a constant rate, along a circular path. In order for this to happen, it is necessary to apply a force of constant magnitude that acts at all times perpendicular to the direction in which the object is moving at that moment. **Figure 1** illustrates the situation at three successive instants of time, t_1 , t_2 and t_3.

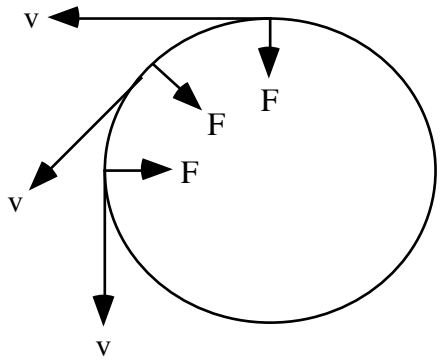


Figure 1: A particle, under the action of a force of constant magnitude that is directed always towards the same point, moves with a velocity of constant magnitude but continuously changing direction, in a circle around that point.

Notice that the magnitudes of both the velocity v and the force F remain constant: Only their directions change with time. Velocity and force are two examples of so-called **vector** quantities, i.e. quantities that have magnitude and direction. This is in contrast to quantities, like mass and time, which have magnitude but no direction. Those kinds of quantities are called **scalar** quantities.

Now we could, in principle, perform an experiment to measure the magnitude of the force F that is responsible for circular motion. We could attach a spring balance (of the kind that measures the *weight*, i.e. forces) to a string of known length r. At the other end of the string we would attach a weight of known mass m. We would then twirl the mass around our head, in a circular path and at a constant rate which we would measure with a clock. A friend would then take a photograph that would enable us to read the force indicated by the spring balance. We could perform many such experiments; some with different weights on the end of the string; some with strings of different lengths; and some at different rates of revolution. When we come to analyse the results we would find that a simple formula relates the mass, string length and rate of rotation to the magnitude of the force. It is:

$$F = m v^2 / r \qquad (4.1)$$

You will notice that this formula is of the type: "force = mass x something", where the "something" is v^2/r. But from Newton's 2nd Law, this must be the acceleration of the mass.

"How strange!" you are probably thinking. The mass moves with a velocity of constant magnitude and yet it is accelerating. But this is not strange. We defined acceleration as the rate of change of the velocity, and in this case the velocity is changing - in direction! If the force had been acting from behind, it would increase the magnitude of the object's velocity (i.e. speed it up). Conversely, if the force had been acting from ahead, it would decrease the magnitude of the object's velocity (i.e. slow it down). If the force had acted from behind at some arbitrary angle to the object's motion, it would both speed it up and change its direction. Similarly, if the force had been acting from ahead but at some arbitrary direction to the object's line of motion, it would both slow it down and change its direction. Clearly then, if the force acts precisely perpendicularly to the line of the object's motion, it would neither slow it down nor speed it up: It would *only* change its direction. This is the case we have for circular motion. For such motion, the acceleration is:

$$a = v^2 / r \qquad (4.2)$$

Just like velocity, acceleration is also a vector quantity. In the present situation the magnitude of the acceleration remains constant but its direction changes continuously. However, at all moments, the direction of the acceleration a is in the same direction as the force F that causes it.

The force F, in eq. (4.1) is sometimes called a *centripetal* force from the Greek for "causing to move in a circle".

2. Centripetal forces and gravitation

Let us now take a look at how knowledge of the centripetal force enables us to make all kinds of interesting astronomical calculations. As our first example, consider a planet of mass m, moving in a circular orbit of radius r about the sun of mass M. From Newton's law of gravity we know that the centripetal force (i.e. the force of gravity that maintains the planet in orbit around the sun) takes the value:

$$F = GMm/r^2 \qquad (4.3)$$

where G is Newton's gravitational constant, which we encountered in the previous lecture.

By equating eq. (4.1) and eq. (4.3) we see that the mass of the planet cancels out and we are left with a relationship between the radius of the planet's orbit and the magnitude of its velocity:

$$r v^2 = GM \qquad (4.4)$$

Now the velocity in orbit is usually not a much-used parameter. Instead, one usually talks about the *period* T of a planet. This is the time it takes to complete one orbit (i.e. the length of a year on that planet). Clearly, the relationship between T and v is:

$$v = 2\pi r / T \qquad (4.5)$$

If we insert eq. (4.5) into eq. (4.4) we obtain the relation:

$$r^3 / T^2 = GM / 4\pi^2 \qquad (4.6)$$

Eq. (4.6) is remarkable in that it applies equally to *any* planet, no matter what its mass m happens to be.

Let us check it for the earth. From astronomical tables, the earth-sun mean radius is r = 1.496 x 10^{11} m, and the average length of 1 year is T = 3.156 x 10^{7} s. Therefore the left-hand side of eq. (4.6) = 3.361 x 10^{18} m^3 s^{-2}. On the other hand, G = 6.673 x 10^{-11} m^3 kg^{-1} s^{-2}, the sun's mass is M = 1.989 x 10^{30} kg. Therefore, the right-hand side of eq. (4.6) = 3.362 x 10^{18} m^3 s^{-2}. The small difference is due to the fact that I have rounded all of the data to 3 decimal places.

Furthermore, eq. (4.6) does not only apply to our sun and its planets. If M is the mass of some distant star, eq. (4.6) enables astronomers to calculate the orbital radius of one of its planets if the period is known. Often, the planet is far too close to its parent sun to enable it be resolved by a telescope but one observes its periodicity via changes in brightness of the light emitted by the star.

Eq. (4.6) also applies to the motion of the moon around the earth and, indeed, also to that of any artificial earth satellite.

3. Fictitious Forces

Suppose we were to hold an empty picture frame in our hands and view the world through it. Everything would appear pretty normal, unless we started to jerk the frame in arbitrary directions. In that situation, the world would appear to be jerking about too - like a movie made by a bad cameraman. If we didn't realize that the frame was jerking and were to try and understand the strange motion of the objects we were looking at, we might arrive at the erroneous conclusion that some kind of weird forces were at work - perhaps an earthquake or something. These forces would, of course, be **fictitious** - purely the result of viewing the world from an accelerating frame of reference.

In the above example I have exaggerated in order to make a point. If, instead of a jerking picture frame, we were to view the earth from the window of an accelerating rocket ship, we could compute the earth's relative motion to us by assuming that a fictitious force was accelerating the earth away from us. We would know that the force *is* fictitious because accelerometers on board our ship (and also our bodies) would tell us that *we* were accelerating, whereas accelerometers on the earth would record nothing unusual down there. Remember that velocity is relative but acceleration is absolute. In order for there to be no fictitious forces we must use a so-called **inertial** reference frame, i.e. one which is not accelerating.

Now let us return to the earth. Because the earth rotates about its axis, it is actually an accelerating reference frame. Fortunately, for most purposes, the resulting fictitious forces are very small. If this were not the case, Newton might never have discovered his famous laws! But there are situations in which the fictitious forces associated with the earth's rotation do become manifest. Two such fictitious forces are of interest to geographers. One is the so-called **centrifugal force**, which appears to act even on stationary objects. The other is **coriolis force**, which becomes apparent when objects move.

3.1 Centrifugal force

This is the easier of the two fictitious forces to understand, because we experience it in everyday life. Each time we round a leftward bend in an automobile we "feel" a fictitious centrifugal force trying to throw us sideways to the right. In fact, what we actually feel is the reactive force of the car wall pushing us to the left when, by Newton's First Law, we are trying to continue in a straight line. Quantitatively, centrifugal force is numerically equal to centripetal force but in the opposite direction.

In our previous discussion of planetary motion around the sun, instead of saying that the centripetal force is the gravitational force, as we did when we derived eq. (4.4), we could have obtained the same result by using the following reasoning: *When a planet revolves around the sun, there are two forces acting on it which must balance each other precisely. One is the force of gravity, pulling it towards the sun. The other is the centrifugal force, throwing it away from the sun.* It doesn't matter which kind of reasoning you prefer to use as they both give the same answers.

Let me give two examples in which the concept of centrifugal force helps provide a simple intuitive explanation for two geographical phenomena. The first, is the shape of the earth, which is not perfectly spherical: While the young earth was cooling down but still molten, its rotational motion about its own axis created a centrifugal force that tried to eject material into space (mostly towards the equator, where the magnitude of the rotational velocity is greatest). The resulting equatorial radius (6377 km) is thus slightly larger than the polar radius (6356 km). This difference in radii is one factor that causes the precise value of g, the acceleration due to gravity, to vary slightly at different places on earth: At the poles, we are nearer to the center of the earth than we would be at the equator.

A second factor that causes g to take a slightly different value at the equator may also easily be understood by thinking in terms of centrifugal force: When we stand at the equator, gravity pulls us down with an acceleration of, say, 9.81 m s^{-2}; but the centrifugal force caused by the earth's rotation tries to throw us out into space with an acceleration of v^2/R, where R is the radius of the earth and v is the tangential velocity of the earth's surface as it rotates on its axis once every 24 hours.

$v = 2\pi \times 6.377 \times 10^6 \text{ m} / (24 \times 60 \times 60 \text{ s}) = 463.7 \text{ m s}^{-1}$

Hence,

$v^2/R = (463.7 \text{ m s}^{-1})^2 / (6.377 \times 10^6 \text{ m}) = 0.03 \text{ m s}^{-2}$.

We see thus, that the earth's rotational motion *reduces* the value of g by 0.03 m s^{-2}, compared to what it would be at the poles.

3.2 Coriolis force

When you sit in a children's merry-go-round it is easy to "feel" the centrifugal force, fictitious as it is. But if you try walking in a straight you will be in for another surprise. In order to see what happens; let us examine the case of a marble being rolled across a rotating turntable. We shall assume that there is no friction between the marble and the turntable, so that the marble does, indeed, travel in a straight line, as demanded by Newton's First Law,

and as would be evident to an observer looking down on the marble from above. **Figure 2** shows three successive instants of time, t_1, t_2 and t_3, starting from the projection of the marble from point A towards point B.

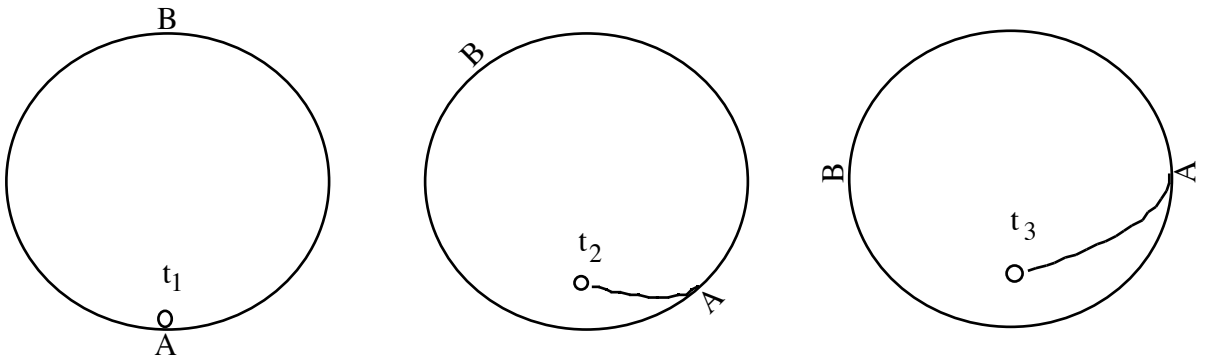


Figure 2: The marble moves in a straight line, at constant velocity, along the direction that AB pointed at time t_1. In real space (as seen by we, who look down upon the turntable) the marble continues along this straight path, see at subsequent instants t_2 and t_3. However, to someone turning with the turntable, the path of the marble appears to be a complicated curve, as shown in the figure. The rotating observer interprets this curved path as being due to a fictitious coriolis force.

In **Figure 2** I have also drawn a curve which indicates the path that the marble would appear to follow if viewed by an observer who was rotating with the turntable. Such an observer would interpret the non-linear path as being due to a fictitious force, which is called the coriolis force.

Unlike the centrifugal force, which is easy to describe quantitatively, the mathematical description of the coriolis force is too complicated to derive in this course. So I shall not write it down. The coriolis force due to the earth's rotation does have a number of physical manifestations in situations where trajectories are very long (e.g. rivers that flow great distances in a north-south direction, or wind regimes), or for situations in which velocities are very high (e.g. for long range military projectiles). However, it is too weak a force to manifest itself on the small scale (e.g. for bathwater running down the drain!).

Problem set 4 (forces and gravity):

1. Communication satellites move around the earth, above the equator, at the same rate at which the earth rotates on its axis. In this way the satellite appears to stand stationary at a fixed point in the sky. Such an orbit is called a **geostationary orbit**. Calculate the height above the earth's surface, in km, of a geostationary satellite. [For this calculation you should take: $G = 6.673 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, the mass of the earth $M = 5.976 \times 10^{24}$ kg, the mean radius of the earth R = 6371 km, and one day = 24 hours].

2. Look up the orbital parameters r and T for all of the planets in the solar system and examine the extent to which eq.(4.6) is valid for each of them.

3. The river Nile flows approximately from south to north, from near the equator to a latitude of approximately 30° N. On which bank would you expect to find more erosion, the left or the right? Explain why.

Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 5, v.4.2 (December 2, 2003)

1. Conservation Of Energy

In lecture 2 we encountered the *Law of Conservation of Momentum*. Although the concept of *momentum* is not as familiar in everyday life as is *velocity*, for the description of motion, its conservation - via the recoil of guns, etc. - is relatively easy to grasp. Velocity, on the other hand, is not conserved: That is why we need to introduce momentum in the first place.

Now, *energy* is also conserved but the *Law of Conservation of Energy* is much more subtle. The reason for this is that energy takes a large variety of forms so it is not always easy to see that it is conserved. Let us, as an example, consider a meteor, traveling through space. Consider all the changes in energy it undergoes.

Far from the earth (and from any other massive body) it has only kinetic energy, $p^2/2m$ and rest-mass energy, mc^2. Or so we might at first imagine! But, as it approaches the earth something strange happens: It speeds up! The meteor's kinetic energy increases but its rest-mass energy does not decrease. So where is this extra energy coming from?

From Newton's Law of Gravitation we understand that the meteor and the earth are attracting one another, and that the nearer they get together the stronger will be the force of gravity. Remember: if we half the distance, we quadruple the force, etc. This force, by Newton's Second Law, produces an acceleration which causes the meteor's velocity to increase. Thus the source of the increasing kinetic energy of the meteor must be some form of gravitational energy.

We can understand this phenomenon in terms of energy, by assigning to the distant meteor a property called *potential energy*, which is simply proportional to its distance from the earth. As the meteor approaches the earth, it loses potential energy and gains kinetic energy at the same rate. Or, if you like, its potential energy is converted into kinetic energy. In this manner, energy is conserved.

Close to the earth's surface it is convenient to measure the potential energy (P.E.) of an object as the product of its *mass m, times the acceleration due to gravity g, times its height above the surface h*. Thus,

$$\text{P.E.} = m\,g\,h \qquad (5.1)$$

If the object is released at rest from a height h, then, as it falls, it will lose potential energy and gain kinetic energy (K.E.). At the moment it reaches the ground, its potential energy will be zero and it will have a kinetic energy that is equal to the original value of the potential energy. Thus,

$$\text{K.E.} = m\,v^2/2 = \text{P.E.} = m\,g\,h \qquad (5.2)$$

Eq. (5.2) enables us to calculate the velocity if the height is known. Notice that the mass cancels out, so any object, no matter how massive, will fall at the same rate and reach the ground with the same speed. This is what Galileo discovered in Pisa!

You may object that this seems a bit arbitrary. After all, we can understand all aspects of the meteor's motion from Newton's Law of Gravitation alone - without ever having to introduce the concept of potential energy. Your objection is correct. But there are three reasons for introducing the concept of potential energy. First: convenience. It is simpler, in many cases - including the present example - to calculate with energies rather than with forces. Second: generality. Not only gravitational forces but also many other types of force in physics (e.g. electrical forces) can be replaced by corresponding potential energies. In each case, greater simplicity is achieved in performing calculations.

The third reason is, perhaps the most amazing of all. In fact, it is so remarkable that it would not usually be introduced to physics students until they study for a more advanced degree. It goes under the title of *Noether's Theorem*. I want to talk a little about Noether's Theorem for two reasons. First, it gives a deeper insight into why these various mechanical concepts I keep throwing at you - in a seemingly arbitrary manner - are not arbitrary. Second, it will be a convenient manner for me to introduce the third and final conservation law you will need to know about.

2. A Philosophical Digression into the Relationship between Conservation Laws and Symmetries in Nature

Before I state what Noether's *theorem* is, I must emphasize that it is very much like a theorem in geometry, in that it can be approached from either side. For example, once you grasp the meaning of Pythagoras' theorem, you realize that not only do right-angle triangles possess a certain, well-known, relationship among the squares of their sides but also, conversely, if any triangle is found to obey this relationship then it *must* contain a right angle. Now for Noether's theorem, which may be simply stated: *Any symmetry of nature implies and is implied by a conservation law*.

In the case of the laws of conservation of momentum and energy the corresponding symmetries turn out to be simple but fundamental properties of space and time - this is why I said that Noether's theorem is "amazing". What fundamental symmetry can we assign to *time*? That it flows forward and backward at equal rates? Nonsense: Time only flows forward! So, to what property of something as vague as time can we assign a symmetry? The answer, if we think about it, is the rate at which time proceeds. Have you ever looked at a clock and wondered how we know that all 5 sec. intervals are truly equal to one another? They don't always "feel" similar! Noether's theorem provides an answer: If time is *homogeneous* (i.e. 5 sec. now is exactly the same as 5 sec. last week) then <u>something</u> must be conserved, and that something turns out to be *energy*. Conversely, if energy is conserved then time <u>must</u> be homogeneous. Proof of this requires a graduate course in Mechanics because it is highly mathematical. But it is such an amazing and simple relationship that I wanted you to know about it, even if you cannot prove it. Now, just as it is possible to grasp Pythagoras' theorem without knowing how to prove it, let me move on to what Noether's theorem implies for the symmetries of space.

Empty space is also *homogeneous*: i.e. 1 m near the earth has the same length as 1 m near Mars. As you may now have guessed, Noether's theorem relates the *homogeneity of space* to

another conservation law. This one is the law of *conservation of momentum*. Once again, the proof is highly mathematical but the concept is simple: If space is homogeneous then momentum must be conserved; conversely, if momentum is conserved then space *must* be homogeneous.

Thanks to Noether's theorem, the mechanical laws of conservation of momentum and energy, may be understood as being simple *requirements* of the fact that space and time are homogeneous. In similar manner, the conservation of electric charge may be understood as being a requirement of a certain kind of symmetry possessed by electromagnetic fields. Physicists use Noether's theorem for other, more puzzling, conserved quantities in order to try and probe the, largely unknown, field equations responsible for the various elementary particles (e.g. quarks and neutrinos) but here we are beginning to stray too far from the needs of a geography course. So let us return to empty space.

A moment's thought will convince you that space, unlike time, has another symmetry property in addition to homogeneity: It is also *isotropic* - i.e. the same in all directions. Noether's theorem therefore requires something to be conserved. That something is called *angular momentum*, which we shall return to below.

At this stage you may well be asking whether there is no limit to the mechanical concepts and jargon that I can throw at you. But, thanks to Emmy Noether, there is such a limit and we have now reached it. Space and time have only three symmetries among them and these imply three and only three conservation laws. Physicists have, as I have already mentioned, found other quantities to be conserved by nature - e.g. electric charge and other properties of the fundamental particles of nature - but these conservation laws have nothing to do with the properties of space and time, and hence lie outside the realm of mechanics. We can thus begin to sense a deep unity that exists among the laws of physics. They are not simply individual expressions of the outcome of unrelated experiments but, rather, a unified description of the way objects *must* behave because of the fundamental properties of the universe they occupy: Truly environmental physics at the deepest level!

3. Back To Energy

After this philosophical digression, let us return to our meteor as it speeds towards the earth. It is losing potential energy and gaining kinetic energy. But then it encounters the earth's atmosphere and a new phenomenon occurs. It begins to encounter molecules of nitrogen and oxygen. To begin with, the effect is barely noticeable since the atmosphere is so thin - it simply pushes these molecules aside. Of course, in the process of transferring some energy to those molecules the meteor will lose some energy. But, as the atmosphere becomes ever denser, the kinetic energy lost by the meteor becomes more and more significant: We say that *friction* is slowing it down. Where is the lost kinetic energy going? It is heating up the surrounding air - i.e. increasing the kinetic energy of the gas molecules. But friction is a two-way process: the surrounding air is also heating the meteor which, at a certain temperature, gets so hot that it starts to emit light and we can see it moving across the sky. It thus starts to lose *light energy*. It becomes what we call a *meteorite*. Let us see what is happening at the atomic level.

Atoms consist of a dense positively charged nucleus of tightly-bound protons and neutrons, surrounded by a lightweight, "cloudlike" distribution of negatively charged electrons. I placed quotation marks around the word cloudlike because for a massive atom, such as lead,

it is not difficult to imagine the 82 electrons behaving like a cloud. But even for hydrogen, it's one and only electron also behaves like a cloud in that it is not possible to know where the electron is at any given moment. This is one of the peculiar lessons of *quantum theory* which describes all phenomena at the atomic scale and smaller. Furthermore, quantum theory tells us that, although we can not know where precisely the individual electrons are located in an atom, they must have one of a discrete set of allowed values for their energy. In the normal state of an atom, all electrons sit in their minimum allowed energy states, called "the ground state energy". However, atoms can become excited to higher states of energy by the absorption of energy from outside. When this happens, some of the electrons have their energies raised to higher excited states among the allowed energy states of that atom. However, such excited states are not stable: each excited electron falls back to its ground state by the emission of photons - i.e. quanta of light. In the case of our meteor, the impacts of the rapidly moving nitrogen and oxygen molecules, as they fly past, impart energy to the electrons of the atoms comprising the meteor and raise them to excited energy states. As those electrons fall back to their ground states light is emitted. That is the light that we see as the meteorite streaks across the sky. In fact, all light is generated this way, whether from a match, an electric lamp, or the sun. In the case of a match, frictional forces excite its electrons; in the case of a lamp, electrical energy does the job; and in the case of the sun, nuclear reactions are ultimately (but indirectly) responsible for the light we see.

It is possible to hit an electron so hard that it leaves its parent atom altogether and the atom remains positively charged. But not for long! It soon grabs an electron from some nearby convenient source. In the case of our hot meteorite the nearby source are the oxygen atoms of the surrounding air. The meteorite thus catches fire and burns. In the process, it loses a certain amount of *chemical energy*.

At this stage the meteorite is losing energy in the form of light and chemical reactions, not to mention pieces that are physically breaking off. Finally, if the original meteor was large enough to start off with, its remains will hit the ground. If you happen to be nearby you will hear the "whoosh" as it flies through the air and the "thud" as it hits the ground. These are examples of *sound energy* - pressure waves excited in the air.

After we have let the meteorite cool down we may well wonder what has happened to all the energy it started with. Well, we now know the answer. The meteor lost energy but that energy was ultimately absorbed by other bodies (both massive, and massless - i.e. photons): none of it disappeared from the universe.

4. Conservation of Angular Momentum

Let us now return to Noether's theorem and the three symmetries associated with space and time. We stated that the homogeneity of space and time lead, respectively to the laws of conservation of momentum and energy. Time is not isotropic (since it does not flow in both directions) so there is no additional conservation law here. But space *is* isotropic and the associated conservation law is the *Law of Conservation of Angular Momentum*.

Now, the isotropy of space means that if you stand still and look around you it looks the same in all directions. Angular momentum is therefore, not surprisingly, a quantity having to do with rotating objects. It's conservation has the effect of preserving the rate at which objects rotate. Moreover, since angular momentum is also a vector quantity, the direction in which their axis of rotation points is also conserved. When I say "the rate at which objects

rotate" I am not referring to their angular velocity because, like linear velocity, this quantity is not conserved. Angular momentum - which we shall discuss more quantitatively below - is the quantity that is conserved. A common example is the gyroscope, which can keep a ship's compass horizontal even in heavy seas. Gyroscopes, of course, gradually slow down because of friction, but in empty space they would spin forever - as, for example, the earth does.

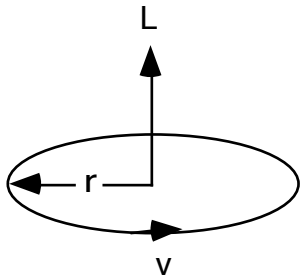
The earth itself is, of course, a giant gyroscope rotating in space at the rate of one rotation each 24 hours. If you imagine holding a mini globe of the earth in your right hand, and turning it in the direction in which your fingers are pointing, then your thumb points in the direction we call "north". This is the way the earth rotates: it is responsible for the sun appearing to rise in the east and set in the west. Don't try this with your left hand or you will get exactly the opposite result!

Now the earth also revolves around the sun, once in approximately 365 days. Interestingly, the rotation of the earth around the sun is also "right handed". So too are the rotation of the other planets of the solar system and the rotation of their moons about those planets that have them. [There are a small number of exceptions that we shall not discuss in detail]. This regularity, together with the law of conservation of angular momentum, suggests that the sun and solar system were formed from a single cloud of primeval matter which broke into smaller pieces each of which continued spinning in the same manner as the original dust cloud. The central piece became the sun whereas the smaller pieces became the planets and their moons.

Now it helps our understanding of angular momentum if we become slightly more quantitative. Suppose we have an object of mass m and velocity v (and hence momentum p = mv) rotating in a circle of radius r about an axis. We define the *magnitude* of the angular momentum as:

$$L = m\ v\ r = p\ r \qquad (5.1)$$

and its *direction* as being along the axis of rotation in the direction of your right thumb! In the figure you see a particle of mass m moving with velocity v in a circular orbit of radius r. The angular momentum has magnitude mvr but its direction is as shown by the arrow L.



This has an interesting and important implication for the earth's motion about the sun, as we shall see in the next lecture.

Problem set No.5 (energy and angular momentum)

[1] A 5-storey building is 15 m tall. It has windows 2 m, 5 m, 8 m, 11 m and 14 m above ground level. Use the law of conservation of energy to calculate the velocity of an object as it passes each of the windows after falling off the roof.

[2] A railway car of mass m, moving with velocity v, strikes a stationary car of mass 3m. If kinetic energy is conserved in this collision (i.e. none of the energy escapes from the collision), use the law of conservation of momentum in order to calculate the speed and direction of each car after the collision.

[3] The moon has a mass of 7.33×10^{22} kg. It orbits the earth at a mean distance of 384,000 km. The average time it takes to complete one orbit is 27.3 days. Calculate its mean velocity in orbit (in m s^{-1}) and the magnitude of its angular momentum (in Js = kg m^2 s^{-1}).

[4] An early (over-simplified) description of the hydrogen atom pictured the electron as orbiting the proton in a similar manner to which the moon orbits the earth. If the electron has mass $m = 9.11 \times 10^{-31}$ kg, orbital radius $r = 2.12 \times 10^{-10}$ m and angular momentum $L = 2.11 \times 10^{-34}$ J s, calculate the magnitude of its orbital velocity v.

Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 6, v.2.3 (December 9, 2003)

1. Motion of the earth-sun system

A *day* is most simply defined as the time it takes the earth to rotate once about its own axis. We can observe this rotation by looking at the stars at night. They appear to rotate about a fixed point in the sky (located very close to the position of the star *Polaris*, if you are in the Northern Hemisphere. The corresponding point about which the stars in the southern sky appear to rotate is a bit more complicated to find as it does not lie close to any of the easily visible stars). Now, the earth also revolves around the sun once in approximately 365 days. This motion can be charted, by studying the gradual night-to-night change in the apparent positions of the stars.

The earth is, of course, held in its orbit by the gravitational attraction between it and the sun. Now here is an interesting situation. In 24 hours the earth will have advanced in its orbit around the sun by 1 day out of a full orbit of 365 days, i.e. by approximately 1º. This means that if we set our clocks to read "noon" the moment when the sun appears directly overhead, we shall have to wait until the earth has rotated approximately 361º before we see the sun in the same position the next day. This is because the orbital motion of the earth around the sun makes the sun appear to be displaced eastwards by about 1º each day. Now this is confusing because it is more convenient to measure time using the sun than the distant stars (the latter being the only "fixed" reference frame relative to which the rotation of the earth can be measured).

In order to overcome this confusion, *two* distinct types of day are defined. The *sidereal day* is a day measured relative to the fixed stars: i.e. a day based on an axial rotation of 360º. Alternatively, the *mean solar day* is one based on the average length of time between the sun being overhead on two successive occasions. Clearly the mean solar day is about 4 minutes longer than a sidereal day.

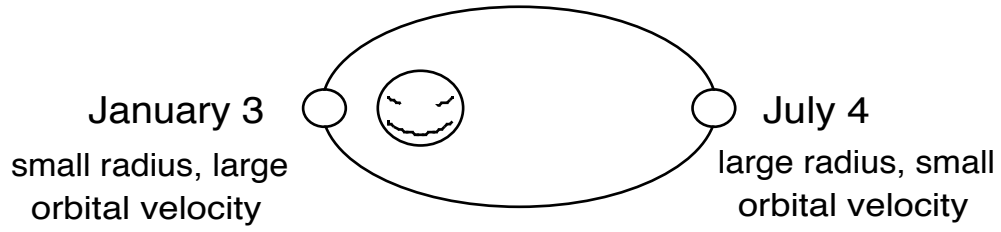


Figure 1: Elliptical orbit of the earth (with exaggerated ellipticity), with the sun at one focus

You will, no doubt, have noticed the adjective "mean" attached to the definition of a solar day. To see why this is necessary let us examine the shape of the earth's orbit a bit more closely. Actually it is not precisely true that the earth rotates around the sun. It is more correct to say that the earth and the sun both rotate about their common center of gravity. However, the mass of the sun is so much greater than that of the earth (333,000 times!) that their common center of gravity lies very close to the sun. The net effect is that we may regard the sun as being essentially stationary and the earth as moving around it in a *slightly* elliptical

orbit. We are in fact closest to the sun on January 3rd and farthest away on July 4th (but the difference is only about 3 %). In **Figure 1** I have exaggerated the ellipticity of the orbit in order to make this point clearer.

However, conservation of angular momentum, as we learnt last lecture, implies that on January 3rd the orbital velocity of the earth must be greatest, and on July 4th it must be least. To see this more clearly, if m is the mass of the earth, v is its orbital speed and r is its distance from the sun then, conservation of angular momentum requires that the earth's orbital angular momentum in January is equal to its value at all other times, for example, in July. Hence:

$$m\, v_{Jan}\, r_{Jan} = m\, v_{Jul}\, r_{Jul} \qquad (6.1)$$

In January, r is small; therefore v must be correspondingly large in order for their product to remain constant.

This speeding up and slowing down of the earth in its orbit implies that one solar day will not be precisely the same length of time as the next. This is the reason we have to define a *mean solar day*. It also implies that simple sundials can not be accurate clocks! This is because if we mark off hourly intervals from noon (i.e. the moment when the sun appears highest in the sky) on one day we shall discover that the next day, our solar clock will be slightly slow or fast depending upon which time of year we calibrated it.

Now there is also another effect that contributes to the difference between one solar day and the next. But before telling you what it is, let me discuss how we describe the location of points on the surface of the earth.

2. Longitude and Latitude

The millimeter graph paper we talked about in the first lecture is useful for plotting graphs on flat surfaces. In fact we plotted such a graph in Lecture 3 in order to discuss velocity, acceleration, etc. But square graph paper is not very useful for plotting graphs on curved surfaces because on such a surface you can walk 1 km south followed by 1 km east and then 1 km north and end up at the same place: Something that is impossible on a plane surface. So, how do we specify locations on a curved surface? In the case of the earth, we draw a set of so-called *great circles*, i.e. circles of maximum possible diameter, that all pass through the north and south poles. *Four* such great circles are shown in **Figure 2**. Can you see them?

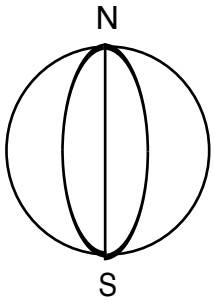


Figure 2: Schematic diagram of the earth, showing *four* great circles

By arranging a grid of evenly spaced circles of this kind we define a set of so-called *lines of longitude*. We measure the angular separation between lines of longitude in degrees. By

convention, and for historic reasons, the 0º line passes through Greenwich, near London, UK. This line is sometimes called the *Greenwich meridian*. Similarly, we talk about locations being east of the Greenwich meridian or west of it. For example, Beersheva has longitude 34.6º E, New York City has longitude 74º W. Note that as you get closer to the poles the lines of longitude all bunch together whereas they are at their maximum separation around the equator.

Exercise: If the earth's equatorial radius is 6,378 km, what is the separation between each degree of longitude around the equator?

Now just as we need both x and y axes on a flat surface we need another set of reference circles on the earth's surface - otherwise we could not distinguish between Mitzpe Ramon, Sede Boqer, Beersheva and Tel Aviv, which all happen to have the same longitude.

By convention, the second set of circles is all chosen to be parallel to the equator. They are called *lines of latitude* and are measured in degrees north of the equator and degrees south of the equator, where the angular measurement is performed relative to the center of the earth. **Figure 3** shows three lines of latitude: the equator (0º by convention), 30º N and 60º S. The figure on the left is the way these lines would appear on a map. The figure on the right is a cross-section through the earth, showing how the angles are measured.

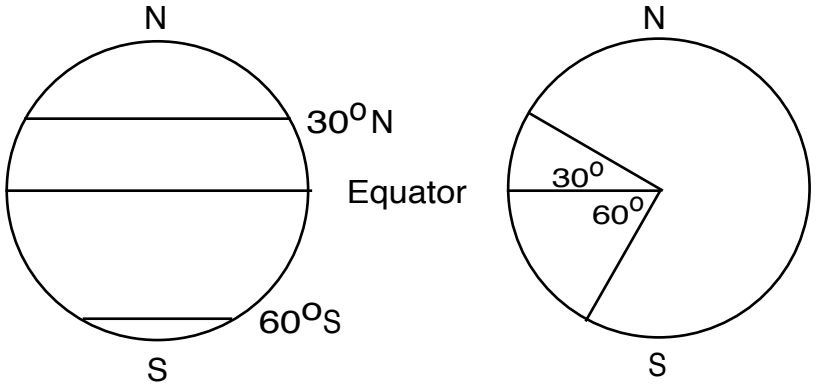


Figure 3: Schematic diagram of the earth showing 3 lines of latitude. Left-hand-side shows the lines as they might appear on a map. Right-hand-side is a cross-section through the earth showing how the angles are defined

Note that, unlike lines of longitude, the lines of latitude do not bunch together. We can now distinguish Mitzpe Ramon as having latitude 30.6º N, Sede Boqer as having latitude 30.9º N, Beersheva as having latitude 31.2º N and Tel Aviv as having latitude 32.1º N. When greater accuracy is required, each degree is subdivided into 60 minutes and each minute is further subdivided into 60 seconds. These are not minutes and seconds of time but of *arc*.

3. Time Zones

In principle, *noon* is the time of day when the sun reaches its highest point in the sky (called "the zenith"). Since this moment will not be the same for two observers located relative to one another along an east-west line, it is necessary to define so-called *Time Zones* for which, by agreement, noon occurs at the same time for all people within the zone. By universal agreement, the 360 degrees of longitude are divided into 24 times zones of 15 degrees (i.e. 1

hour in time) each. Wintertime in London, England, is referred to "Universal Time" (UT). As we proceed eastward in 15-degree steps from London we encounter, successively, the time zones: UT + 1 hour, UT + 2 hours, UT + 3 hours, etc. On the other hand, as we proceed westward from London we pass through the time zones UT - 1 hour, UT - 2 hours, UT - 3 hours, etc. Because the earth is round, the times zones UT +12 and UT - 12 have the same local time. How broad is a time zone? In theory it is precisely 15 degrees but in practice it is a matter of local policy. For example, most of continental Europe, including Spain, has chosen to be in time zone UT - 1 even though in the case of Spain, the UT time zone would be more natural.

There is one further practical complication associated with time zones. It concerns what day of the week it is. Suppose it is midnight between Saturday night and Sunday morning in London. Then in time zone UT + 12 it should be noon on Sunday. On the other hand, in time zone UT - 12 it should be noon on Saturday. But it cannot be Saturday and Sunday at the same time! Correct. In order to overcome this problem an *International Date Line* is defined. This coincides mainly with the longitude = 180º line, which passes mainly over water in the Pacific Ocean. In those few places where the 180º line passes over land, the International Date Line is made to deviate from it so that the latter passes only over water along its entire length. Then, by definition, as you pass across this line from east to west, you *lose* one day. For example, if it is Saturday just east of the line, then it is the same time Sunday just west of the line. On the other hand, if you cross the line from west to east, you gain a day: Sunday becomes Saturday which you can live all over again!

Two final points regarding the local definition of time: (1) Some countries move their clocks forward by one hour (and sometimes more) in summer time in order to save fuel. This is sometimes called "daylight saving time", or simply "summer time". (2) Some countries (e.g. India) subdivide the universal time zones into half-hour sub-divisions. Thus, for example, when it is 10 am in Tel Aviv, it is 13:30 pm in New Delhi.

4. Tilt of the Earth's Axis

Let me now return to that other effect I warned you about which contributes to variation of the length of a solar day. Although the earth rotates about its own axis (once every 24 hours) in the same general direction that it orbits the sun (in approximately 365 days) - i.e., both rotations are anti-clockwise when viewed from above the North Pole - these two rotations are not precisely parallel to each other: the earth's spin axis is tilted at approximately 23.5º to the axis of its rotation about the sun.

This tilt is, of course, responsible for the seasons of the year: It is summer over that part of the globe that is tilted towards the sun and winter over the part that is tilted away from the sun. In the Northern Hemisphere mid-summer occurs around June 21 and mid-winter around December 21. In the Southern Hemisphere the opposite holds true.

But there is another interesting effect of the earth's axis being tilted. In both mid summer and mid winter the two motions responsible for the difference between sidereal and solar time (i.e. orbital motion of the earth around the sun and the apparent motion of the sun from one line of longitude to the next) are precisely parallel to each other. On the other hand, at the so-called *equinoxes* (approximately March 21 and September 21), these two motions have an angle of 23.5º between them. Therefore, just as a change in ocean current will cause a swimmer, swimming at constant velocity relative to the water, to experience a change in his

velocity relative to the shore, so too will the difference between sidereal and solar time vary between these pairs of seasons.

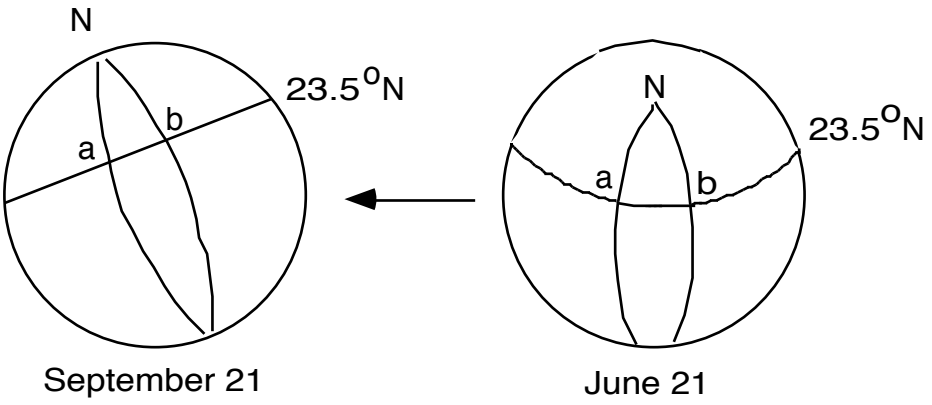


Figure 4: Two schematic diagrams of the earth as viewed from the sun. (left, September 21; right, June 21). The arrow indicates the direction of the earth's orbital motion. The longitude lines passing through points a and b on the earth's surface are separated by 15 degrees

Figure 4 illustrates this situation on June 21 and September 21. In summer a rotation of the earth from a to b represents 1 hr of sidereal time. During this time the earth moves in the direction of the arrow. An observer at position a would see the apparent motion of the sun as being the sum of the earth's orbital and rotational motions. A precisely similar situation would occur in mid winter (i.e. December 21). In autumn however, the sidereal motion of the earth - by precisely the same amount as in summer - is now at an angle of 23.5º relative to the orbital motion and, hence, their combined effect will be slightly less than in summer and winter.

These two effects: that of the ellipticity of the earth's orbit with its annual cycle, and that of the tilt of the earth's axis with its 6-month cycle, together, account for the daily difference between sidereal and solar time. This difference is sometimes referred to as the *equation of time*. The difference between sidereal time and solar time varies from approximately -15 minutes in mid February to approximately +16 minutes at the end of October.

5. Precession of the Earth's Axis

You will probably have noticed that most times you spin a coin the axis of rotation does not remain fixed in a vertical direction. Instead, it moves in a conical manner that we call *precession*. Those toy gyroscopes do the same thing when placed on the miniature Eiffel Towers they come with. The phenomenon of precession is well understood but requires a fairly elaborate mathematical formalism for its quantitative description. Since such formalism lies outside the scope of this course I'll tell you in words what is happening:

If you consider a coin that is standing on its edge but leaning over slightly, it will obviously fall over. Why? Because a gravitational force pulls its center of gravity downward while the table pushes upward on the lowermost part of the coin. Since these two opposing forces do not act along the same line they will obviously *rotate* the coin onto one of its sides, as in the right hand side of **Figure 5**.

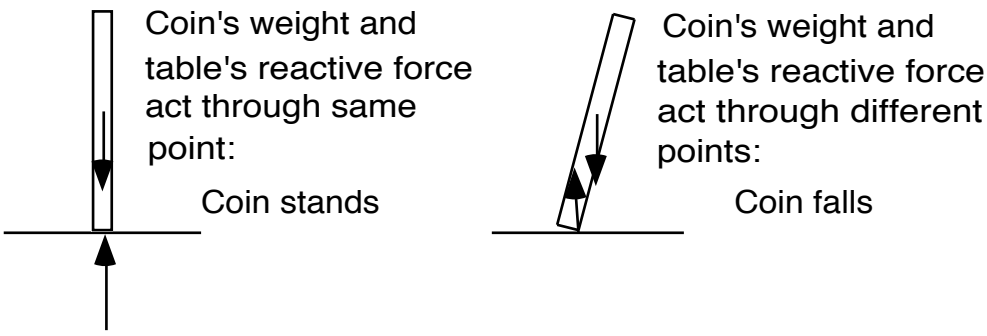


Figure 5: Sideways view of a coin standing on its edge. Left: The coin stands because its weight and the table's reactive force act through the same line. Right: The coin falls over because these two forces do not act through the same line - Their combined effect rotates the coin about a horizontal axis

Now spin the coin. A moment's thought will enable you to realize that there are now *two* rotation effects happening simultaneously: one trying to spin the coin about a vertical axis while the other tries to rotate it about a horizontal axis tangential to the bottom edge of the coin. Now we already encountered the law of conservation of angular momentum - which causes rotating objects to remain with the *direction* of their rotation axis fixed. But what happens when an object - like a coin - has two competing rotation axes? The answer is that these two rotation axes combine to form a third effective axis and it is the direction of this latter axis that is conserved by the law of conservation of angular momentum.

This is not as strange as it may seem. You are all familiar with having to allow for a side wind when throwing a ball, or having to allow for an ocean current while swimming or sailing. This allowance, mathematically, is referred to as the combination of *vectors*. There is one vector that describes your velocity relative to the water and another vector that describes the velocity of the water relative to the land. The vector that describes your velocity relative to the land is simply the combination of the first two vectors.

Well, rotations are also vectors: the combination of two of them (using an appropriate mathematical rule which we need not bother about in this course) produces a third effective rotation. Returning now to our spinning coin: If the coin is spinning fast and almost vertically then the dominant rotation will be its spin. In this case its precession will take the form of a very narrow cone. The more tilted it is, however, at the moment you let it go, the larger will be the rotation that is trying to topple it. In this case the resulting precession will be a much wider cone.

We can now talk about the earth. As we already saw, the earth's rotation about its own axis is not precisely perpendicular to the plane of its orbit around the sun - it is tilted at 23.5^{o}. It therefore precesses. You might wonder *why* it precesses: After all, it is not standing on a table and the gravitational force of the sun seems to act through the earth's center of gravity. So where is the additional rotation coming from? The answer is that, unlike a coin, the earth does not have a perfectly regular geometric shape and its mass is not uniformly distributed. (We encountered this feature of the earth when we discussed the force of gravity and changes in the value of g over different parts of the globe). As a result, the gravitational attraction of the sun does not act precisely through the earth's geometrical center. This is a very small effect but it is enough to cause the earth's axis to precess on a time scale of one rotation in about 26,000 years!

Problem set 6 (the earth and its motion)

1. Where on earth would you be standing if you could walk 1 km southward, then 1 km westward, then 1 km northward, and end up where you started?

2. There is more than one answer to the above question. How many more can you find?

3. The planet Venus has an equatorial radius of 6,050 km. What is the distance between each successive degree of longitude around the equator of Venus?

4. Use maps with a scale 1:100,000 and estimate the latitude and longitude (in degrees and minutes) of the following sites in Israel: Kibbutz Malkiya (Upper Galilea), Kibbutz Giv'at Brenner (Central Plain), Kibbutz Ein Gedi (Northern Aravah), and Kibbutz Yotvatah (Southern Arava).

5. If it is 10 am Tuesday in Oakland, California (Approx. longitude = 122° W), what time and day is it in Auckland, New Zealand (Approx. longitude = 175° E)?

Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 7, v. 3.2 (December 16, 2004)

1. Electrostatic Forces

In Lecture 4 we studied Newton's law of gravity. You will recall that it involves force with the mathematical form:

$$F = G m_1 m_2 / r^2 \qquad (7.1)$$

where m_1 and m_2 are the masses of the two bodies which attract one another (measured in kg), r is the distance between them (measured in m), and G is Newton's gravitational constant (= 6.67×10^{-11} m^3 kg^{-1} s^{-2}). Note that G takes the same value whether the masses are small, as is the case of molecules of air in the earth's atmosphere, or large, as in the case of the earth and the sun.

Another kind of force that occurs in nature was discovered by Coulomb. It is the force between two electrically charged objects. Remarkably, Coulomb's law for electric charges has a similar mathematical form to Newton's law of gravity. It is:

$$F = K q_1 q_2 / r^2 \qquad (7.2)$$

where q_1 and q_2 are the amounts of electric charge (measured in special units called "coulombs", C), r is the distance between the charges (measured in m), and K is Coulomb's electrostatic constant (= 8.99×10^9 kg m^3 s^{-2} C^{-2}). The introduction of electric charges into the simple world of mechanics requires the use of a new "dimension" in addition to mass, length and time. That dimension is electric charge. However, if you check the overall dimensions on the right hand side of eq, (7.2), you will see that the C's cancel out and we are left with kg m s^{-2}, which are the correct dimensions for force.

There is, however, one important difference between Newton's and Coulomb's laws. Because masses can only be positive numbers, Newton's force is always an *attractive force*. However, electric charges can be positive or negative. Therefore, Coulomb's law can be *attractive or repulsive*. If both charges have the same sign, the force is repulsive; if they have opposite signs the force is attractive. For example, two electrons repel one another, whereas an electron and a proton, which have equal but opposite charges, attract one another. This latter fact allows the hydrogen atom to exist. For, the hydrogen atom is something like the earth and moon, in that a lightweight electron "orbits" around a massive proton. In fact, an early model of the hydrogen atom, constructed by Niels Bohr, pictured the electron as circling in an orbit exactly as the moon circles the earth. Bohr equated the coulomb attraction between the two particles to the centripetal force on the electron.

2. Bohr's Model of the Hydrogen Atom

In Lecture 4 we equated the gravitational force between a planet and the sun to the planet's centripetal force. If we do the corresponding thing for the electron and proton in a hydrogen atom, we have:

$$K\,q^2 / r^2 = m\,v^2 / r \qquad (7.3)$$

where q is the magnitude of the charge on either the electron or the proton = 1.60×10^{-19} C (they have identical charges - just opposite signs), r is the distance between them, m is the mass of the orbiting electron = 9.11×10^{-31} kg, v is the speed of the electron along its orbital path. This leads, as in the case of gravity, to a fixed relationship between v and r, no matter how far from the proton the electron orbit happens to be. Namely:

$$r\,v^2 = K\,q^2 / m \qquad (7.4)$$

In the case of gravity, that is as far as we can go. Namely, *whatever value is taken for r*, the value of v will be given by the gravitational equivalent of eq. (7.4). However, Bohr discovered that there is another restriction for atomic systems - and it is an extremely peculiar one. Bohr realized that the orbital angular momentum of the electron could *only* take one of the *discrete* values:

$$L = m\,v\,r = n\,L_0, \quad n = 1, 2, 3, ... \qquad (7.5)$$

where $L_0 = 1.05 \times 10^{-34}$ J s is a constant known as "Planck's constant divided by 2π". This is peculiar because we would have expected that L could take any value whatsoever. But nature has decided otherwise. L can be L_0, $2L_0$, $3L_0$, etc., but nothing in between.

Planck's constant is the underlying numerical constant behind so-called *quantum mechanics*; that is the mechanics that describes the way molecules, atoms and other small-scale systems interact with one another. One of the unfortunate aspects of modern physics is that there is no known simple way to picture quantum mechanics - but it is one of the most accurate known computational tools in all of physics. Even Bohr's model, useful as it certainly is, gives nonsense if we try to dig too deeply into it in terms of a physical picture. For example, if we eliminate v between equations (7.5) and (7.4), we find that the distance between the electron and proton can *only* take one of the *discrete* values:

$$r = n^2\,L_0^2 / (K\,m\,q^2), \quad n = 1, 2, 3 \qquad (7.6)$$

where *all* symbols on the right hand side of this equation are constants. This implies that the electron can not be at any arbitrary distance from the proton; only at discrete allowed distances, given by the values of the so-called *quantum number* n. However, what happens when an electron jumps from one orbit to another (which it *can* do)? Surely, our senses tell us, it must pass through the forbidden region. But if it does, then the region is not forbidden!!! This is just one example of the conceptual problems we get into if we try and picture in our imagination what is actually happening in the quantum world.

Similarly, the kinetic energy of the electron can also *only* take discrete values:

$$(1/2)\,m\,v^2 = m\,K^2\,q^4 / (2\,n^2\,L_0^2) \qquad (7.7)$$

where, again, everything on the right is constant.

And the electrostatic potential energy is also *quantized* (i.e. it takes only discrete values):

$$-K\,q2 / r = -m\,K^2\,q^4 / (n^2\,L_0^2) \qquad (7.8)$$

You will notice that the potential energy is negative, and twice the magnitude of the kinetic energy. Therefore, the total energy of the electron, being simply the sum of the kinetic and potential energies, is also quantized:

$$E = - m K^2 q^4 / (2 n^2 L_0^2) \qquad (7.9)$$

Note that the *sum* of the kinetic and potential energies of the electron in a hydrogen atom is *negative*. This means that the mass of a hydrogen atom is *slightly less* than the sum of the masses of a proton and an electron. What do I mean by "slightly"? The rest mass energy of a proton is 938 MeV (where MeV is a convenient unit of energy), the rest mass energy of an electron is 0.511 MeV, but the largest value of E, in eq. (7.9) is only 13.6×10^{-6} MeV.

3. Waves

Everyone is familiar with the phenomenon of waves in water. If you drop a stone into a pond of still water, a circular wave spreads out from the point where the stone fell. Now, the stone originally contains potential energy before it is dropped. That energy is converted to kinetic energy as the stone reaches the pond, and most of this kinetic energy is transferred to the water molecules. It is this energy that causes the wave to move.

Now a water wave is an interesting example of what is known as *collective motion*. As the wave spreads out from the stone's point of impact, all that happens is that water molecules move up and down locally. No water actually travels with the wave: it only looks that way. The wave, however, carries energy with it. As it spreads out, each part of the wave contains less and less energy. Hence, as time passes, the wave is less and less able to move water molecules up and down. The wave thus, eventually, dies out.

Suppose now, instead of dropping a single stone, we arrange for a motor-driven hammer to strike the water repeatedly. This time a series of waves spread out from the hammerhead. If the hammer strikes the water at a constant frequency, ν times per second, the waves will spread out with a fixed separation λ between successive wave crests. It is easy to see that the speed v with which the waves move through the water is:

$$v = \nu \lambda \qquad (7.10)$$

We refer to λ as the *wavelength* of the wave, and ν as the *frequency* of the wave. For example, if ν = 5 crests per second (which is denoted 5 Hz, or simply 5 s^{-1}), and λ = 10 cm, then from eq. (7.10), the wave spreads out at a speed v = 50 cm s^{-1}.

In the case of water waves, wave motion is easy to understand, but there is another kind of wave that was discovered in the 19th century, the *electromagnetic wave*, that was much more difficult to understand. The reason is that electromagnetic waves can travel through a vacuum!

Electromagnetic waves have wavelength and frequency, just like any other kind of wave. Those with wavelengths of hundred of meters are familiar as *AM radio* waves. Those with wavelengths of about 1 m are familiar as *FM radio* and *Television* waves. Those with wavelengths of about 10 cm are familiar as *microwaves*. Those with wavelengths of about 10 μm are familiar as *infrared radiation*. Those with wavelengths in the range 400 - 700 nm are familiar as *light waves*. Those with wavelengths in the range 250 - 400 nm are familiar as

ultraviolet radiation. At shorter wavelengths still, we encounter first *X-rays*, then *gamma rays*. All of these are examples of electromagnetic radiation.

Strangely, it was quantum mechanics that helped clear up the mystery of how electromagnetic waves can travel in empty space. The reason for this is that although in the large-scale world around us, particles and waves are completely different objects, at the atomic scale this is no longer the case. At the atomic scale, particles can behave like waves and waves can behave like particles. Richard Feynman, one of the pioneers of our modern understanding of quantum mechanics called them all *wavicles*. How does this help us understand the ability of electromagnetic waves to travel through a vacuum? Well, if an electron can travel through a vacuum, so too can a *photon*. A photon is the particle manifestation of electromagnetic waves. But then, does an electron also have wave properties? It certainly does - otherwise we could not have electron microscopes.

Quantum mechanics gives us simple relationships between the particle-like and wave-like properties of these wavicles: Wavelength of the wave is related to momentum of the particle by de Broglie's equation:

$$p = h / \lambda \qquad (7.11)$$

and frequency is related to energy by Einstein's equation:

$$E = h \nu \qquad (7.12)$$

where h is a constant known as Planck's constant ($h = 2\pi L_0 = 6.63 \times 10^{-34}$ J s).

Notice that if we divide eq. (7.12) by eq. (7.11) we obtain:

$$E/p = \nu \lambda = v \qquad (7.13)$$

i.e. the speed of the wave is the ratio of the corresponding particle's energy to its momentum. In Lecture 2 we learned that, being a massless particle, the photon has E = pc, where c is a constant equal to the speed of light in a vacuum. Eq. (7.13) thus tells us that all photons (no matter what their wavelength) travel at the speed of light.

And what about electron waves? The situation is slightly more complicated here. Eq. (7.11) remains true, but eq. (7.12) must be modified slightly. If you plug eq. (7.11) into the energy-momentum-mass relationship in Lecture 1, you will discover that instead of E, the left hand side of eq. (7.12) now becomes $\sqrt{(E^2 - m^2c^4)}$. As a result, the velocity of electron waves also turns out to be c, even though the electrons themselves move at slower speeds. But here we are departing rather far from "environmental physics" so it is time to stop.

Problem set 7 (hydrogen atoms and electromagnetic waves)

1. (a) How many atoms are there in 1 kg of hydrogen gas? (b) What is the total mass of all the protons? (c) What is the total mass of all the electrons [1 molecule of hydrogen contains 2 atoms; the proton mass = 1.67×10^{-27} kg, the electron mass = 9.11×10^{-31} kg].

2. Insert numbers into eq. (7.6) in order to calculate the radii of the first Bohr orbit (n = 1).

3. If the hydrogen atom were bound by a gravitational force rather than an electrostatic force, how large would its first Bohr orbit be?

4. A hydrogen atom in an excited state n = 2, returns to its ground state (n = 1) by emitting a photon. Assuming that energy is conserved, calculate the wavelength of the photon.

Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 8. v.6.1 (December 23, 2003)

1. Larger atoms than hydrogen

In the previous lecture we saw how a hydrogen atom can exist in various energy states, specified by a quantum number n which may take any positive integer value: 1, 2, 3, ... etc. For each of these allowed energy states we calculated that the total energy of the electron (i.e. kinetic plus potential) takes the value:

$$E = - m K^2 q^4 / (2 n^2 L_0^2) \qquad (8.1)$$

where m is the electron mass, = 9.11×10^{-31} kg, K is Coulomb's electrostatic constant, = 8.99×10^9 kg m^3 s^{-2} C^{-2}, q is the magnitude of the electric charge of the electron, = 1.60×10^{-19} C, L_0 is the fundamental unit of angular momentum, = 1.05×10^{-34} J s (better known as "Planck's constant divided by 2π"), and n is the integer which tells us which energy level we are talking about.

We also learned that the energy of a photon is simply related to the wavelength, λ, of the corresponding light wave, via the relationship:

$$E = 2\pi L_0 c / \lambda \qquad (8.2)$$

where c is the velocity of light in a vacuum, = 3.00×10^8 m s^{-1}, and λ is measured in meters.

This simple mathematical model provided a qualitative *and* quantitative explanation of the spectrum of light that was observed from evacuated electrical discharge tubes in which trace amounts of hydrogen were present. The explanation was that:

1. Energetic photons in the electric field within the tube (which had obtained their energy from the external electric power supply) collide with electrons in the ground states, n = 1, of the hydrogen atoms, and excite them to one of their higher energy states, $n = n_i$.

2. Hydrogen atoms in excited states are unstable and get rid of their unwanted energy by emitting photons.

3. The energy carried off by each photon is precisely the amount which enables an atom to de-excite to one of its less excited states, until, ultimately, it is in its ground state, n = 1.

4. It is the discrete wavelengths of these emitted photons that are observed by the spectroscopists.

5. For an atomic de-excitation from state $n = n_i$ to a state $n = n_f$, (where $n_i > n_f$), eq. (8.1) tells us that the hydrogen atom will lose an amount of energy equal to:

$$E_i - E_f = [m K^2 q^4 / (2L_0^2)] (1/n_f^2 - 1/n_i^2) \qquad (8.3)$$

And from eq. (8.2), the emitted light will have a wavelength:

$$\lambda = 2\pi L_0 c / (E_i - E_f) \qquad (8.4)$$

For other kinds of atom the story is similar but, not surprisingly, more complicated. The reasons for the additional complication are: (1) All other atoms, being more massive than hydrogen, contain more than a single electron, and those electrons interact with one another, in addition to interacting with the positively charged atomic nucleus; (2) It turns out that further quantum numbers (in addition to the number n which we encountered for hydrogen) restrict the allowed energies of the various electrons. Let me give two examples.

My first example is *helium*, the second lightest atom (denoted by the symbol He). The He nucleus contains 2 protons and is surrounded by 2 electrons. Incidentally, you may think that a helium atom has twice the mass as an atom of hydrogen, but, in fact, it has 4 times the mass. This is because its nucleus contains 2 other, electrically neutral, particles called *neutron*s. The neutron and proton have almost precisely the same mass as one another. Fortunately, the presence of neutrons in the nucleus does not complicate the "chemistry". I.e., they do not effect the motion or energies of the electrons.

Now, the allowed energy levels of helium are qualitatively quite similar to those of hydrogen, as you will discover when you solve homework problem 1 for the case of *singly-ionized* helium (where one electron has been removed, so as not to complicate the calculation). Here we have an electron of charge -q that is attracted to a nucleus of charge +2q. The result is that each helium energy level n has 4 times the magnitude of the corresponding hydrogen level. Other than this overall energy scale difference, the results are qualitatively similar for the two kinds of atom. However, helium has 2 electrons in its natural (i.e. un-ionized) state. Therefore, you might guess that in the ground state of helium, both electrons would sit in the n = 1 energy state (or the n = 1 *orbital*, as chemists call it). And you would be correct. The fact that there are two electrons does not disturb the energy levels very much. There would naturally be a tendency for the two negatively charged electrons to be as far apart from one another as possible within the n = 1 orbital. But otherwise there are no particular surprises for this atom.

My second example is *lithium*, the third lightest atom (denoted by the symbol Li). The lithium nucleus contains 3 protons and usually 4 neutrons. This atom therefore contains 3 electrons. Now, you might guess that when a lithium atom is in its ground state, all three of the electrons would be in the n = 1 orbital. But this is not the case. Only 2 of its electrons occupy the n = 1 orbital; the third electron resides in the n = 2 orbital. Why is this the case?

It turns out that because of further quantum properties of electrons, that I shall not discuss here, a maximum of 2 electrons are allowed in the n = 1 orbital. Any further electrons have to reside in the higher orbitals staring with n = 2. But the n = 2 orbital can only contain up to a maximum of 8 electrons. If the atom contains more than 10 electrons (i.e. 2 in the n = 1 orbital and 8 in the n = 2 orbital), then the remaining electrons have to reside in the n = 3 orbital, but again, only up to a certain allowed maximum, above which, any further electrons have to sit in the n = 4 orbital, etc.

2. Interactions among atoms: molecules

If this sounds complicated, I apologize, but that's the way it *is*. In fact, I have actually over-simplified (!) this description in order to make it seem as simple and plausible as possible.

Perhaps a diagram will help. **Fig. 1** shows a table of the atoms which have all of their electrons in the first 3 orbitals. The first row, H and He, are those atoms whose electrons occupy the n = 1 orbital. The second row, Li - Ne, are those atoms with full n = 1 orbitals and, successively, from 1 to 8 electrons in their n = 2 orbital. The third row, Na - Ar, are atoms which have full n = 1 and n = 2 orbitals, and which have electrons in their n = 3 orbital. Those of you having a familiarity with chemistry will recognize **Fig. 1** as being the first 3 rows of the famous *Periodic Table of the Elements*, discovered empirically by Mendeleev in the 18th century. Mendeleev created this table by studying the way in which chemical compounds react with one another. For example, the column of gases on the extreme right (He, Ne, Ar,...) have no chemical reactions with other elements. Chemists call them "The noble gases". On the other hand, those elements in the column immediately preceding these noble gases, (F, Cl, ...), known as "the halogens", react very strongly with the column of elements (H, Li, Na, ...), known as "the alkali metals", which immediately follows the noble gases.

1 H							2 He
3 Li	4 Be	5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar

Figure 1: The first 3 rows of the Periodic Table of the Elements

Mendeleev could not understand the atomic basis that causes chemical reactions to exhibit these regularities, but we can. It turns out that atoms with all orbitals full are stable, in the sense that they have minimum energies. These are the noble gases. Because they already exist in minimum energy states, they can not reduce their energies by entering into chemical reaction with other elements.

On the other hand, look at the situation of fluorine (F) in the halogen column. This element has 7 electrons in its n = 2 orbital. If it could "find" another electron it too would be energetically stable. But ... now look at the alkali metals column. Lithium has only 1 electron in its n = 2 orbital. If it could get rid of that electron it too would be energetically stable (having a complete n = 1 orbital). Thus, the situation is perfect for lithium and fluorine to "do a deal". Li donates an electron to F. In this way, they are both energetically more stable. However, in the process, Li becomes positively charged (having lost an electron) and Li becomes negatively charged (having gained an electron). They solve this problem by "sticking together" to form LiF, thereby preserving charge neutrality. In this way, a *molecule* of LiF has lower energy than a pair of free Li and F atoms. It is consequently more stable.

But chemical reactions can also occur between 2 atoms of the same kind. Take the case of hydrogen. As two free hydrogen atoms approach one another, each "thinks" that if it could capture the other's electron, it could complete its own n = 1 orbital. So they too "do a deal": Some of the time one atom has both electrons and some of the time the other atom does. Here too, the possession of both electrons renders the *acceptor* atom negatively charged and the *donor* atom positively charged. They therefore stick together, forming an electrically neutral

hydrogen molecule. Again, the energy of a hydrogen molecule is lower than that of two free hydrogen atoms.

3. Atomic number, ionization, atomic mass and isotopes

A number of terms that are commonly used by chemists are easy to understand using the simple picture of atoms we have described above. First, *atomic number* Z is simply the number of protons in the nucleus of an atom. It is also equal to the number of electrons if the atom has not lost any by *ionization.* Atomic numbers are, obviously, always integers: for H, Z = 1; for He, Z = 2; for Li, Z = 3, etc.

What is ionization? If an atom is hit by a sufficiently energetic photon, one of its electrons can be removed completely. The resulting atom is then said to be "ionized" An ionized atom is positively charged so it can not remain free for long before regaining the lost electron in some way or other (usually by combining with another atom that has an "extra" electron - or, rather, one that can be borrowed).

The *atomic mass* A is the number of particles, i.e. protons and neutrons, in the nucleus. Simple hydrogen has atomic mass A = 1 because its nucleus contains only one particle, a proton. However, hydrogen can also exist in two heavier forms: *deuterium*, which has a neutron added to its nucleus; and *tritium*, which has 2 neutrons in its nucleus. These three forms of hydrogen are called *isotopes* of hydrogen. They all have the same chemical reactions because they all have only one electron, i.e. Z = 1. However, deuterium has atomic mass A = 2 and tritium has atomic mass A = 3. Water that contains deuterium atoms is sometimes known as "heavy water".

Now, if you consult a "more serious" edition of the periodic table of the elements, than that shown in **Fig.1**, you will discover that atomic masses are generally not integers. For example, boron (B) has atomic mass A = 10.8. How can this be? All this means is that boron occurs naturally in nature in a number of isotopes. Some boron atoms will have 5 protons and 5 neutrons, and others will have 5 protons and 6 neutrons. There may also be small admixtures of other boron isotopes. The value 10.8 is simply the *average* atomic mass of boron.

4. The size of atoms and their constituents

From the previous lecture we might expect that the approximate shape and size of a hydrogen atom would be a sphere of radius:

$$r = n^2 \ L_0^2 / (K \ m \ q^2) \qquad n = 1, 2, 3, ... \qquad (8.5)$$

If you plug in numbers you will find that this turns out to be 5.29×10^{-11} m for the n = 1 (ground) state, four times as large if the electron is in the n=2 energy state, etc. In the more modern quantum theory of matter, the so-called "first Bohr radius" takes on a more complex interpretation than the simple "radius of a circle" as we have developed it in this course. However, its value represents the approximate size of all atoms, in so much as "size" is a useful concept.

In the case of atoms with higher mass values than hydrogen, it is easy to see that for an ionized atom containing Z protons in its nucleus and a single orbital electron, the first Bohr radius takes the value:

$$r = n^2 L_0^2 / (K m Z q^2) \quad n = 1, 2, 3, \ldots \qquad (8.6)$$

Therefore, for a given value of n, the "size" of the atom decreases as Z increases, provided that the number of electrons in the inner orbital shells remains constant.

But what about all of those protons and neutrons in the nucleus? The nucleus and its constituents are several orders of magnitude smaller than the atom. Protons and neutrons have sizes that are typically of order 10^{-15} m. Hence, even if you pack a few hundred of them into a small ball so that they touch one another, the resulting ball will not be much larger than about 10^{-14} m. Thus it is the orbital electrons that determine the size of an atom – not its nucleus.

Of course, the more protons you try to pack into such a small volume, the stronger will be their electrostatic repulsion against one another. Very heavy nuclei will consequently tend to lose some protons, spontaneously - they are said to be *radioactive*.

You are probably wondering how it is possible to place even 2 protons in an atomic nucleus (as in helium) and stop them from blowing each other apart. The answer is that, in addition to electrostatic interactions, protons also possess so-called *strong interactions*. Strong interactions, as their name suggests, are stronger than electrostatic interactions, and attractive. They have a different mathematical from than the Coulomb interaction, and one which is extremely short range. Two neutrons, which also possess strong interactions but no electrostatic interactions, would not begin to feel each other until they approached within a distance of about 10^{-14} m of one another. Electrons, on the other hand, possess electrostatic interactions but no strong interactions. Consequently they can not feel neutrons. Apart from holding the constituents of atomic nuclei together, strong interactions play no role in nature at large. We therefore don't need to know anything about how the protons and neutrons behave within the nucleus. All we *do* need to know, in order to understand nature in its large-scale manifestations (i.e. chemistry, solids, etc.) is how the electrons behave.

Problem set 8 (atoms and molecules)

1. Derive a formula for the energy levels for an atom of singly-ionized helium (in a similar manner to which we derived the Bohr formula for hydrogen atom energies). Calculate the numerical values of the first 3 energy levels (i.e. for n = 1, 2, 3) for this atom. [Note: You may picture an atom of singly-ionized helium as consisting of a single electron orbiting a nucleus which contains 2 protons and 2 neutrons].

2. Calculate the wavelengths of the light from photons that are emitted in the (n = 3 -> n = 1), and the (n =2 -> n=1) transitions in an atom of singly-ionized helium. Compare these results with the corresponding wavelengths for photons emitted by a hydrogen atom.

3. As you move from left to right along the 3rd row of the Periodic Table in **Fig.1**, would you expect the size of the atoms to increase or decrease? Explain your reasoning.

4. Look at the positions of hydrogen (H), carbon (C), and oxygen (O) in the table shown in **Fig.1** and explain *in words*, why the molecules CH_4 (methane), H_2O (water) and CO_2 (carbon dioxide) may be expected to be particularly stable.

Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 9. v.3.5 (December 30, 2003)

1. Solids

We have already seen some examples of how the forces among electrons are responsible for the chemical reactions between atoms. These forces also determine the *average* distance between atoms or molecules in bulk matter, and their relative orientations. I have emphasized the word "average" here because the molecules are not stationary in a solid: They actually vibrate very rapidly about their average positions. I shall say more about this vibration later but, for the present, let us concentrate on the average positions of the molecules.

In the first lecture we encountered the fact that crystals contain regular arrays of atoms spaced typically about 0.5 nm apart. Now, depending upon the specific atoms in question (and external conditions such as temperature and pressure), their geometrical arrangement within a crystal will take one form or another from among a relatively small number of possibilities. Four examples of so-called *unit cells* for crystals are shown in **Fig. 1**. You will notice that each of these cubic unit cells is slightly different from the others.

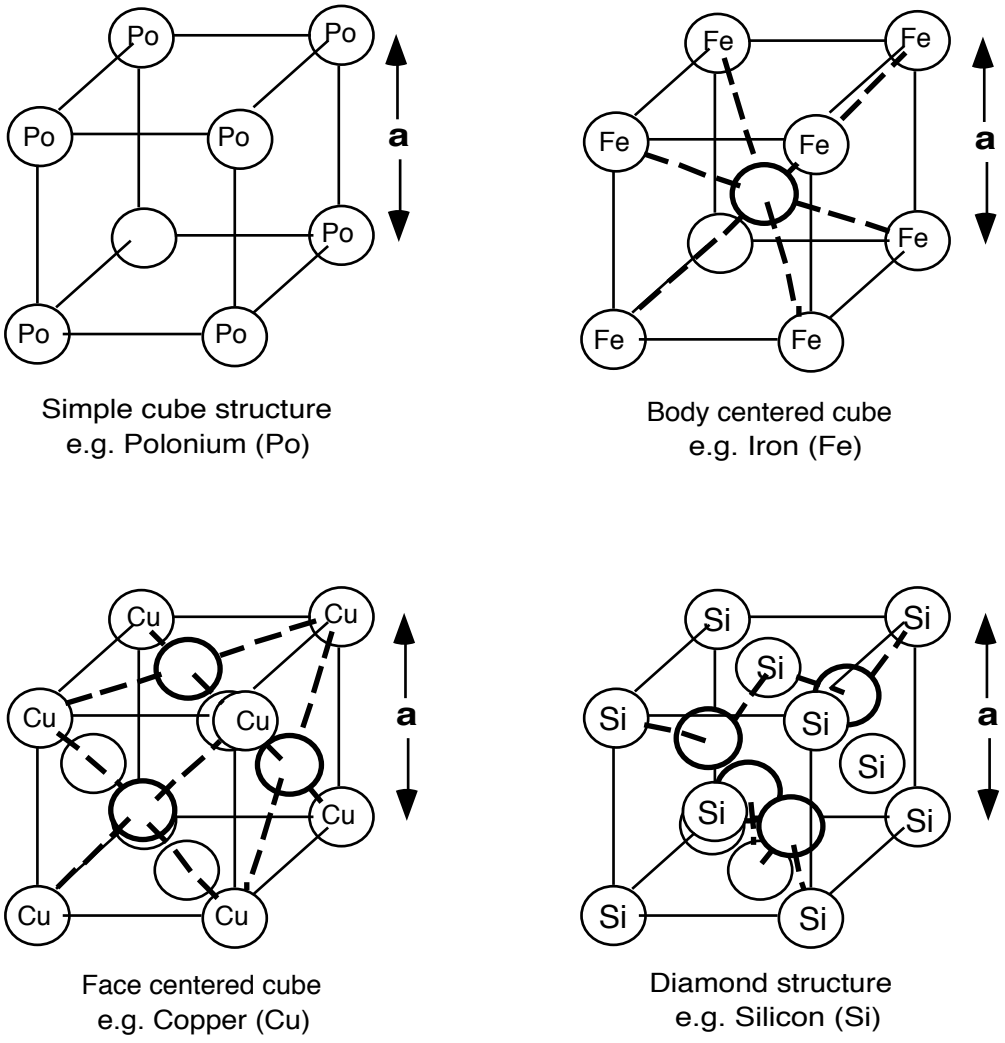


Figure 1: Four examples of possible cubic arrangement of atoms in solids

In the case of the element polonium (Po is element number 84 in Mendeleev's Periodic Table of the elements) the atoms occupy the corners of an empty cube of side **a** (**a** is called the *lattice constant*). This structure is called *simple cubic* and abbreviated *sc*. The upper left diagram in **Fig. 1** shows a unit cell of solid polonium.

Atoms of iron (Fe is element number 26 in Mendeleev's Table) also arrange themselves at the corners of a cube but there is an extra atom in the center of the cube. This structure is therefore called *body-centered cubic*, or *bcc*. The upper right diagram in **Fig. 1** shows a unit cell of solid iron.

Atoms of copper (Cu is element number 29 in Mendeleev's Table) arrange themselves at each corner of a cube and also at the center of each of its faces. Therefore this structure, not surprisingly, is called *face-centered cubic* or *fcc*. The lower left diagram in **Fig. 1** shows a unit cell of solid copper.

Finally, silicon (Si is element number 14 in Mendeleev's Table) atoms arrange themselves in an fcc structure but, in addition, there are 4 extra atoms completely within the cube. This structure - you will probably not guess - is called *diamond structure* or *ds* (because the carbon atoms that make up diamonds take this same crystalline form). The lower right diagram in **Fig. 1** shows a unit cell of solid silicon.

There are also a number of other crystalline shapes, which are not cubic. For example, aluminum crystallizes with a *hexagonal* shaped unit cell, but I shall not discuss these more complicated structures because they require more than a single lattice constant to specify their geometry. I would, however, like to emphasize that *all* minerals crystallize into one of a relatively small number of possible crystal shapes, which relate many of their physical properties to the underlying atomic structure. The following section will illustrate one example.

2. Density of Crystals

An important practical feature of unit cells is that we may consider a real crystal of, say, a few cm in size, as being made up of a large number of unit cells replicated some 10^8 times in each direction. In such a crystal, a corner atom in any given unit cell is actually shared by the 8 unit cells that meet each other at that atom. Similarly, a face-centered atom in a given unit cell is shared by that cell and the adjacent unit cell: i.e. by 2 unit cells. Finally, a completely enclosed atom in a body-centered unit cell belongs only to the unit cell in which it sits.

Thus, a unit cell of a bcc material such as iron actually contains, on average, only two atoms: one atom in the center and 1/8 of an atom at each of its 8 corners.

We may now perform the following simple calculation:

Let the mass of a single atom be m. Then the mass of a bcc unit cell is:

$$M = 2m \qquad (9.1)$$

Since the lattice constant is a, the volume of a bcc unit cell is:

$$V = a^3 \qquad (9.2)$$

Therefore, the density ρ of a bcc unit cell is:

$$\rho = M/V = 2m/a^3 \qquad (9.3)$$

At this stage, just to check that you have understood the argument, you may like to deduce the corresponding expression for the densities of unit cells of type sc, fcc and ds. The results will be useful in the exercises below.

Now we can measure the density of iron by dividing the mass of a piece of the material by its volume (obtained, for example by immersing it in a fluid). The result is 7.86 g cm^{-3}, or in SI units: 7,860 kg m^{-3}.

On the other hand, from our knowledge of atoms, we can calculate the mass of a unit cell of iron since it contains 2 iron atoms, each of which has an atomic mass of A = 55.8, i.e. the mass of each iron atom is 55.8 times heavier than a hydrogen atom (which, as we have seen, weighs 1.67 x 10^{-27} kg). This knowledge therefore enables us to calculate the average distance between the iron atoms!

Re-writing eq. (9.3) as:

$$a^3 = 2m/\rho \qquad (9.4)$$

and inserting numbers, we have:

$a^3 = 2 \times 55.8 \times 1.67 \times 10^{-27}$ kg / 7,860 kg $m^{-3} = 23.7 \times 10^{-30}$ m^3

Hence, by taking the cube-root, a = 2.87 x 10^{-10} m = 0.287 nm, which is the desired answer. That is to say, we have used the physical density of iron, some information from chemistry, and our knowledge of its crystalline structure to calculate the distance between neighboring atoms in the crystal!

Now let us look at what assumptions have gone into deducing this result. First, there is the assumption that solid materials in general, and iron in particular, are made of atoms.

Secondly we have assumed that we know the crystalline structure of iron. This can also be determined experimentally, by scattering X-rays or neutrons from a sample of iron and studying the resulting patterns made on photographic film. In fact, by this technique, one can independently measure the 0.287 nm spacing between iron atoms.

Finally, we made use of some atomic properties of iron atoms which were determined by chemists about 100 years ago, i.e. how many times heavier than hydrogen is an atom of iron and how heavy is a hydrogen atom.

From a detailed knowledge of the forces between atoms, physicists are able to calculate what kind of crystal structures should occur in nature and, also, many of their physical properties (in addition to the density). For example, carbon crystallizes both as a diamond structure - which enables its extreme hardness to be understood - and, also, in the form of graphite. We have not discussed the graphite structure here but its geometrical form allows us to

understand why carbon in this crystalline form acts as a lubricant. An exciting "new" form of carbon was discovered in recent years, when it was realized that the same inter-atomic forces that can produce diamond and graphite can also produce a giant molecule of 60 atoms arranged like the surface of a soccer-ball. This material was named Buckminsterfullerene after the American architect, Buckminster Fuller, who suggested this kind of structure for making large, rigid, dome-shaped buildings.

3. Thermal Energy and Temperature

I'd now like to return to my earlier remark that the geometrical arrays in crystal unit cells represent only the *average* position of the vibrating atoms. The vibrations are caused by the kinetic energy of the atoms. This energy can be increased by providing heat from outside, or decreased by removing heat (via a refrigerator).

As we cool a crystal the atomic vibrations become smaller and smaller, until ultimately, we could think of the atoms as being completely motionless. At that stage it would not be possible to extract any further energy from them, as they would have none to give. Correspondingly, it would not be possible to reduce their temperature any further. Their temperature at that stage would thus truly be *absolute zero*. This temperature turns out to be -273.15 ºC on the *Celsius scale*. It is also denoted 0 K on the so-called *Kelvin scale* - which is just like the Celsius scale except shifted by 273.15 degrees.

If we move in the other direction - by adding energy, i.e., by heating the crystal, the atomic vibrations will become more and more violent. At a certain stage of heating, the amplitude of the atomic vibrations will be larger than the average distance between the atoms. When this happens, the crystal will lose its shape - the solid will become a liquid!

If we continue heating the liquid, some of the molecules will jump right out and never return. At this stage the liquid has boiled and become a vapor!

Again, depending upon the specific atoms and their interatomic forces, the solid, liquid and vapor phases of their existence will occur at different temperatures and pressures. For example, among the halogens, at room temperature: iodine (I) is solid; bromine (Br) is liquid; and chlorine is a gas. If we heat them they all become gases: If we cool them they all become solids.

3.1 Specific Heat (*Chom Seguli*)

The *specific heat* of a material is defined as the *rate of change with temperature of its energy per unit mass*. In the case of liquid water, the specific heat C = 4.19 kJ kg^{-1} K^{-1}. For example, if we wish to know how much energy is required to heat the contents of a 120 liter tank of water from 20^0C to 60ºC, the answer is:

120 kg x 40 K x 4.19 kJ kg^{-1} K^{-1} = 20.112 MJ (= 5.59 kWh) (9.5)

3.2 Latent Heat of Fusion (*Chom Kamus shel Hituch*)

Ice, the solid form of water, has a specific heat of C = 2.10 kJ kg^{-1} K^{-1}. This means, for example, that we would need to add 2.1 kJ of energy in order to heat a 1 kg block of ice from a temperature of -3 ^{0}C to -2 ^{0}C; another 2.1 kJ to raise its temperature to -1 ^{0}C and yet another

2.1 kJ of energy to raise the temperature of the ice to 0 ^{0}C. At this stage something different happens. If we add another 2.1 kJ of energy, the temperature of the ice will remain at 0 ^{0}C and nothing will seem to have happened. But as we continue to add more energy to the ice block, something is actually happening to the molecules inside. The kinetic energy we are supplying to them is allowing them to break loose from the rigid crystalline structure within which they were frozen. Gradually, as we add more energy, the ice melts. After we have added 334 kJ we find that the 1 kg block of ice has completely melted, and we are left with 1 kg of liquid water, still at a temperature of 0 ^{0}C. From then onwards, as we continue to add heat energy, we shall find that each additional 4.19 kJ of energy will raise the water temperature by 1 ^{0}C. The energy needed to melt a sold, L_f = 334 kJ kg^{-1} in the case of ice, is called its *latent heat of fusion*.

3.3 Latent Heat of Vaporization (*Chom Kamus shel Iduy*)

A corresponding phenomenon occurs when a liquid is on the point of boiling: It requires *extra* energy to enable the molecules to leave the liquid - i.e. for the liquid to convert to a gas. In the case of water, this *latent heat of vaporization*, as it is called, amounts to L_v = 2260 kJ kg^{-1}. Thus, 4.19 kJ of energy will suffice to raise 1 kg of liquid water from a temperature of 99 ^{0}C to 100 ^{0}C, but the water will remain in its liquid phase, without increasing in temperature, until we have added an extra 2260 kJ of energy. At that point, all of the water will have converted to steam. Incidentally, if the 1 kg of steam were not lost but collected in a vessel at atmospheric pressure, we would find that each further 1.95 kJ of energy would raise its temperature by 1 ^{0}C, because the specific heat of water vapor at atmospheric pressure is C = 1.95 kJ kg^{-1} K^{-1}.

Problem set 9: (solids and liquids)

1. Obtain the corresponding values for the lattice constants of polonium (atomic weight = 209, density = 9,400 kg m^{-3}), copper (atomic weight = 63.5, density = 8,933 kg m^{-3}) and silicon (atomic weight = 28, density = 2,329 kg m^{-3}).

2. Use the information given in section 3 of this lecture to calculate how much energy is required in order to convert 0.5 kg of ice at -15 ^{0}C into steam at +115 ^{0}C.

Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 10, v.1.4 (January 13, 2004)

1. Liquids

In the previous lecture we saw how the addition of heat energy to a solid may cause its molecules to vibrate with such large amplitudes that the solid loses the geometrical shape that was imposed by the inter-molecular forces. In other words: the solid becomes a liquid.

The temperature at which a solid turns into a liquid depends on the kind of atoms it is made of and the strength of the forces that bind them together. At room temperature, lead (Pb) and gold (Au) are solids but bromine (Br) and mercury (Hg) are liquids. Lead can be easily melted by heating it to 327 °C. Gold, on the other hand, will not melt unless its temperature is raised above 1,064 °C. Rock in the earth's interior is so hot that it is permanently in liquid form. Occasionally this liquid escapes to the surface through fissures in the earth's crust and we see it in this form as a volcanic eruption. As the resultant lava cools off it solidifies again.

One can actually see evidence for liquids being made up of atoms (or molecules) in constant motion. More than a century ago it was discovered that if pollen grains are suspended in water and viewed under a microscope they execute a rapid motion in randomly changing directions, known as Brownian motion (after the botanist Robert Brown who first reported the phenomenon in 1828). Albert Einstein's first famous scientific paper was a quantitative explanation of Brownian motion, which he proved, was caused by the continuous bombardment of the pollen grains by the vibrating molecules of the liquid. Einstein actually used the phenomenon of Brownian motion to demonstrate (for the first time ever) that liquids are made of atoms (or molecules).

Now in order to contain a liquid it is only necessary for the vessel to have a bottom and sides: Not a lid. This is because the upward motion of most of the molecules in a liquid can not overcome a force called *surface tension* - that apparent "film" that seems to hold the surface together. We can actually use our molecular picture to understand the origin of surface tension. Molecules in the bulk of the liquid are acted upon, equally, by attractive forces from molecules above and below. But molecules on the surface can only be attracted by molecules below them: they are consequently pulled by a net downward attractive force. Occasionally, of course, a surface molecule happens to be given enough of a push, by the random collisions among the underlying molecules, to overcome surface tension and cause it to escape. In this manner the liquid gradually evaporates.

In the 3rd Century BCE Archimedes discovered an important property of liquids that remains true to the present day. He observed that *a solid object sinks in water until the upward force it experiences is precisely equal to the weight of water displaced by it.* Let us look at **Fig. 1** to see what this means.

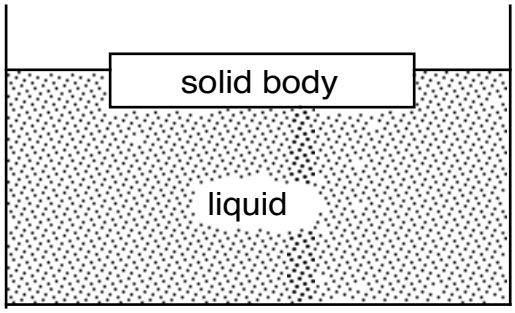


Figure 1: A solid body can float on the surface of a liquid

Fig. 1 shows a slab of solid material, e.g. wood, floating in a liquid, e.g. water. Suppose the contact area between the slab and the water is A and the body sinks to a depth d. Then the volume of water displaced must be V = A x d. Now what has happened to this displaced water? Clearly it must have squeezed upward around the solid body. It would "like" to go back to its original level but it can not do so because the wooden slab is keeping it out. What is the force with which this displaced water is pushing down in order to try and return? Clearly, with a force equal to its weight = A x d x ρ x g, where ρ is its density and g is the acceleration due to gravity. But this force must be precisely canceled by the weight of the slab = M x g. This then is the meaning of *Archimedes' Principle*.

Now Archimedes' Principle illustrates another interesting and important property of fluids: The weight of the displaced water acts downward but pushes the slab upward. How is that possible?

With fluids it is convenient to talk about *pressure*, which is *force per unit area*. The weight Mg of the slab acts on an area A thus exerting a pressure Mg/A on the water. This is the pressure with which the water - which has been squeezed up and around the slab - must push back in order to hold the slab in a stable position.

This shows us that unlike force, which acts in a definite direction (force is a vector quantity), the pressure in a liquid acts uniformly in all directions. Again, from the atomic viewpoint, this is sensible because the pressure is caused by colliding molecules moving very rapidly in random directions.

2. Melting icebergs.

Let us now use these ideas in order to calculate something that the newspapers tell us has profound consequences for the "environment". I refer to the effect on sea level that may be caused if all the icebergs melt. We may simplify the calculation by making an iceberg of cylindrical shape, 10 cm long and 1 cm^2 in cross-sectional area. It thus has a volume of 10 cm^3. Let us place this iceberg in a glass of water at 0°C (so that it will not melt until we are ready).

Now the density of ice is 0.92 gm cm^{-3}. Our iceberg accordingly has a mass of 9.2 gm. However, according to Archimedes' Principle, the iceberg will sink until it has displaced 9.2 gm of water - which, being water (ρ = 1), occupies precisely 9.2 cm^3 of volume. If we mark the water level on the side of the glass, *after* we have floated our mini iceberg, our apparatus may be regarded as representing a simulation of the present level of the oceans with all of their floating icebergs. We are now ready to perform a global warming experiment.

We have a 9.2 gm iceberg, which has displaced 9.2 cm^3 of water. If we raise the water temperature in the glass, the iceberg will melt. How much volume does a 9.2 gm fully melted iceberg occupy? The same volume as 9.2 gm of water, i.e. 9.2 cm^3. But this is *precisely* the volume in the glass that was previously occupied by the submerged part of the solid iceberg.

We conclude, therefore, that if global warming causes all the icebergs to melt there will be no change in sea level!

That is the good news. The bad news is that the Antarctic contains vast quantities of ice that are not floating: this ice is attached to solid ground. If it melts and pours into the sea, it will certainly cause sea level to rise.

3. Tides

Before leaving the topic of liquids I'd like to discuss the phenomenon of tides and why there are two of them every 24 hours. In ancient times it was realized that the moon was related to the phenomenon of tides but it was not until people (a) realized that the world was round and (b) knew about gravitation, that they were able to address the question of how many high tides there should be in 24 hours - one or two. After all, as the earth rotates on its axis once every 24 hours,

the moon should cause an attractive force on the ocean, and we might expect there to be one high tide each day. But there are two! How come?

There was much confusion on this issue until Newton showed how to perform the calculation correctly. The problem was: Given that the moon attracts the sea, why is there also a high tide on the *other side* of the world?

Newton's law of gravitation states that two masses attract each other with a certain force that we discussed in a previous lecture. However, if - as is the case for the earth and the moon - the two masses also rotate around each other, there will also be a centrifugal force trying to separate them. In equilibrium, both bodies will revolve around their mutual center of gravity in such a manner that the gravitational attraction exactly balances the centrifugal repulsion. Let us look at a schematic diagram of the earth-moon system, in which the ocean is included.

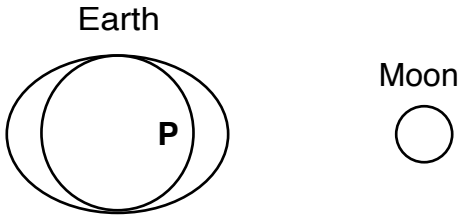


Figure 2: The earth and moon revolve around their mutual center-of-gravity at point **P**. The gravitational force of the moon is greater to the right of **P** and causes a high tide there. The centrifugal force is greater to the left of **P** and causes a high tide there

In **Fig. 2**, the point **P** marks the center of gravity of the revolving earth-moon system. Point **P** is the center around which the entire system revolves and at which the gravitational force precisely balances the centrifugal force. Now the ocean on the side of the earth nearest the moon will, on average, "feel" the gravitational attraction of the moon more strongly than the centrifugal force of rotation - because it is nearer to the moon than the other water. There will therefore be a high tide on the side of the earth facing the moon. This was obvious even without Newton's help! Around the other side of the earth, however, the situation is the opposite. The water here will, on average, feel the centrifugal force more strongly than the gravitational force of the moon. There is accordingly a high tide around the back of the earth, caused, not by an excessive gravitational force but by an excessive centrifugal force. This is what Newton explained.

4. Gases

If we add yet more kinetic energy to the molecules of a liquid the surface tension film breaks apart and the molecules all escape. Boiling has occurred and we shall lose the material completely unless it is contained within a suitable vessel.

A suitable vessel for containment might be a gas jar with a lid. But we can also use the force of gravity (as we shall show later in a more quantitative fashion). If a planet is not very massive then any atmosphere it might have had when it was formed will rapidly have evaporated into space. However, if the planet is massive enough (e.g. the earth) then the force of its gravity is sufficient to trap the atmosphere and stop all but the lightest gas molecules from escaping.

Now, just as some materials can be solid and others liquid at room temperatures, some can be gases. Thus, oxygen, nitrogen and carbon dioxide are gases, water and oil are liquids and coal, iron and silicon are solids.

Quantitative experiments performed on gases, some 300 years ago, revealed a number of interesting regularities between the temperature, pressure and volume of a gas.

First, experiments were performed on gases held at *fixed temperature*. It was discovered that if the pressure is increased the volume decreases and vice-versa. This is almost obvious but what

was remarkable about the measurements was that the product PV remained constant. (This is known as *Boyle's Law*).

Another interesting result was that if the *pressure is kept constant* then heating causes the temperature and volume to increase in simple proportion to one another. provided that temperature is measured on the *absolute temperature scale*, i.e. in degrees Kelvin. (This is known as *Charles' Law*). For example, increasing the temperature by any given factor would cause the volume to increase by that same factor.

Finally, if the *volume* of gas in an enclosed container was *kept constant*, it was discovered that the addition of heat to the container caused the temperature and pressure to rise in simple proportion to one another - again, provided that the Kelvin scale of measurement is used for temperature. (This law was discovered by *Gay-Lussac*).

The results of these three sets of experimental observations have been combined into what is known as the *Ideal gas law*:

$$PV = RT \qquad (10.1)$$

where P is the gas pressure measured in N m^{-2}, V is volume measured in m^3, T is temperature measured in K (Remember: degrees Kelvin = degrees Celsius + 273.15) and R is a constant, called the *ideal gas constant*, which takes the value 8.31 J K^{-1}.

To see that eq. (10.1) gives us each of the proceeding three laws it is sufficient to hold one variable fixed and see how the remaining two depend upon one another:

First, by holding the temperature fixed at some value T_o, eq.(10.1) gives:

$$P V = (R T_o) \qquad (10.2)$$

If V_1 and V_2 are the volumes occupied by a gas at pressures P_1 and P_2, respectively, then, from eq.(10.2) we have:

$$P_1 V_1 = (R T_o) \qquad (10.3)$$

and

$$P_2 V_2 = (R T_o) \qquad (10.4)$$

Equating eq.(10.3) and eq.(10.4) then gives us:

$$P_1 V_1 = P_2 V_2 \qquad (10.5)$$

which is the usual form of Boyle's Law.

On the other hand, if we now hold the pressure constant at some value P_o then eq.(10.1) gives:

$$V = (R/P_o)\, T \qquad (10.6)$$

Again, by considering two states of a gas, in which V_1 and V_2 are the volumes it occupies at temperatures T_1 and T_2, respectively, we obtain:

$$V_1 / V_2 = T_1 / T_2 \qquad (10.7)$$

which is the usual form of Charles' Law.

Finally, if we hold the volume fixed at some value V_o then eq.(10.1) gives:

$$P = (R/V_o)\, T \qquad (10.8)$$

from which we may derive

$$P_1 / P_2 = T_1 / T_2 \qquad (10.9)$$

which is the usual form of Gay-Lussac's Law.

The ideal gas law (also Charles' Law and Gay-Lussac's Law) is actually only an approximate law of nature. The reason for this is that if we could reduce the temperature while holding the pressure fixed then the volume should decrease to zero. But this is impossible since all the molecules would need to be compressed to a mathematical point. This is the sense in which the gas laws (10.1) - (10.9) are said to be *ideal*. But they work remarkably accurately provided that the gas density is sufficiently large that there is no danger of the molecules getting into permanent contact with one another. But this is a reasonable requirement because, as we have seen, long before this happens our gas would have become a solid.

Although the ideal gas law was discovered empirically it has to do with the collisions among gas molecules. We should therefore not be surprised to discover that eq.(10.1) contains important information about the molecules themselves. For this reason it often re-written in the form

$$PV = NkT \qquad (10.10)$$

where, N is *Avogadro's Number* ($N = 6.02 \times 10^{23}$ mole^{-1}) and k is *Boltzmann's constant* ($k = 1.38 \times 10^{-23}$ J °K^{-1}).

Boltzmann's constant is actually a scaled-down form of the ideal gas constant $R = 8.31$ J °K^{-1} in eq.(10.1). The scale factor - Avogadro's Number - represents the number of molecules contained in a so-called *mole* of any substance. A mole is the gram molecular mass of a substance, i.e. the number of grams of a substance that are numerically equal to its molecular weight. For example, a mole of solid carbon (C) weighs 12 gm because the molecular weight of carbon is 12: it contains $N = 6.02 \times 10^{23}$ carbon atoms. Similarly a mole of liquid water (H_2O) weighs 18 gm (because the atomic weight of oxygen is 16 and that of hydrogen is 1): it contains $N = 6.02 \times 10^{23}$ water molecules. Similarly, a mole of oxygen *gas* (O_2) weighs 32 gm: it contains $N = 6.02 \times 10^{23}$ oxygen molecules. Note that a mole of oxygen *atoms* would weigh 16 gm, but the atoms would quickly combine in pairs leaving us with only half-a-mole of stable oxygen gas.

But you are probably wondering what all this has to do with an ideal gas, for which no specific atoms have been mentioned. Well, the answer is that (3/2) kT turns out to be the average kinetic energy of a gas molecule at temperature T, irrespective of what kind of molecule it is. Hence, although we originally defined it in terms of an ideal gas, the right hand side of eq.(10.10) actually represents 2/3 of the kinetic energy of all N gas molecules in our volume V - whatever kind they may be. In this sense we see that temperature is proportional to the kinetic energy of the molecules in a gas.

Problem set 10 (liquids and gases)

1. The element mercury is a liquid at room temperature, with density = 13.55 gm cm^{-3}. A copper disk with density = 8.9 gm cm^{-3}, diameter = 2 cm and height = 1 cm is placed on the surface of a pool of mercury. To what depth will the copper disk sink? Draw a sketch diagram with all relevant dimensions labeled.

2. An iceberg is made of pure water ice (density = 0.92 gm cm^{-3}). It is floating in salty sea water (density = 1.03 gm cm^{-3}). If the iceberg melts, will the sea level rise or fall?

3. If the moon had the same mass as the earth, would there be 2 tides or 1 every 24 hours? Explain.

4. If dry air consists of 78% nitrogen, 21% oxygen and 1% argon, what is the mass of 1 mole of air? How many molecules of each type does it contain?

Environmental Physics for Freshman Geography Students

Professor David Faiman. Lecture 11, v.4.1 (January 20, 2004)

Atmospheric phenomena

In this final lecture we shall address a number of physical phenomena that occur within the earth's atmosphere and their effect on the environment in which we live. The major constituents of the air we breathe are: Nitrogen (N_2) 78%, Oxygen (O_2) 21% and Argon (Ar) (1%). These percentages have been rounded to whole numbers. Had I used 3-figure accuracy, their sum would have been 99.9%, the remaining 0.1% containing "trace" amounts of other gases. The most important trace gases – from the environmental view point – are: water vapor (H_2O), of varying fraction, depending on geographic latitude and time of year; and carbon dioxide (CO_2), about 0.03%. These gases and their effects will be discussed below.

1. Stratification of the atmosphere

As we move away from the earth's surface, the density of the air decreases, until eventually we find ourselves in the vacuum conditions of space. However, the *temperature* of the atmosphere does not vary smoothly with altitude. Instead, there are a number of distinct layers, as shown schematically in **Fig. 1**. The principal layers are: the troposphere, the stratosphere, the mesosphere, and the thermosphere.

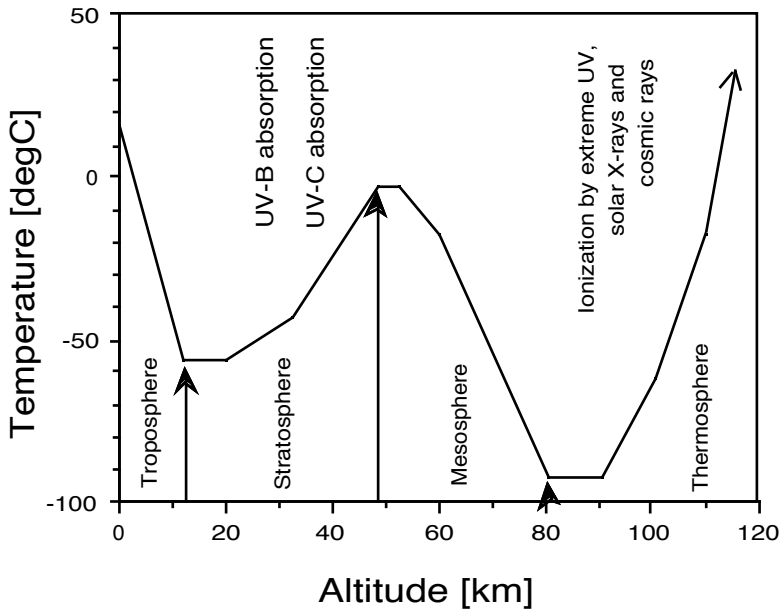


Figure 1: Typical temperature variations with altitude in the troposphere, stratosphere, mesosphere and thermosphere of the earth's atmosphere

1.1 The troposphere (approx. 0-12 km)

The part of the atmosphere that is in contact with the ground is called the *troposphere*. Here the temperature *decreases* with increasing altitude, from ambient values of typically +20 ºC at ground level, to approximately -60 ºC at a height of about 12 km. At this altitude we encounter a transition region known as the *tropopause*, where the air temperature stops falling. The precise values of the tropospheric temperature vary from summer to winter, and the height of the tropopause is different at the equator compared to the poles. But the values given in **Fig. 1** are typical representative values.

The physical reason for this fall in temperature with increasing altitude in the troposphere is that air is basically transparent to solar radiation. Most of the incoming solar radiation is absorbed, *not by the air*, but in the ground, which becomes heated as a result. The warm ground then heats the nearby air via a process of *conduction*, i.e. via the transfer of kinetic energy from the vibrating ground molecules to the less energetic air molecules. Each warmed layer of air then passes on some of its energy to the next layer, etc. However, the further we get away from ground level, the weaker will be this conductive heating because the energy will have been shared among more and more air molecules. For this reason, the temperature of the air will fall the higher we go in altitude. However, as already mentioned, after about 12 km, this cooling stops and something else starts to happen.

1.2 The stratosphere (approx. 12-50 km)

At the height of the tropopause we notice that the temperature of the air stops falling, and then starts to rise! Here, we have entered the *stratosphere*, where the temperature now *rises* the higher we go, reaching almost 0 ºC at about 50 km above ground.

The agent responsible for heating in the stratosphere is the kinetic energy generated by solar *ultraviolet light* (UV). As we shall discuss below, these photons are so energetic that they can break oxygen molecules into free oxygen atoms. It is the energy absorbed from these photons and converted to kinetic energy that causes the atmospheric constituent molecules to heat up within the stratosphere. At the top edge of the stratosphere, we encounter a second transition region, known as the *stratopause* where the air temperature stops rising. Again, this must indicate some new physics.

1.3 The mesosphere (approx. 50-80 km)

By the time we reach the stratopause there are very few oxygen molecules left in the atmosphere. Therefore, solar UV radiation ceases to be an effective heating agent because there are not enough molecules to absorb these energetic photons. We have reached the *mesosphere*, where the atmosphere once more starts to cool off with increasing altitude. Here the temperature drops to nearly -100 ºC at an altitude of 80 km where, at a third transition region - the *mesopause* – the physics changes yet again.

1.4 The thermosphere (above approx. 80 km)

After the mesopause we enter the *thermosphere*, where the air pressure has dropped to 10^{-12} mb. In fact there are so few gas molecules per unit volume that "temperature" loses its normal meaning. However, there are now several agents that can raise the energy of these residual

gas molecules; namely, extremely short-wave solar UV radiation, X-rays and cosmic radiation. If one translates the resultant kinetic energy of the gas molecules into "temperatures", using (3/2)kT as the kinetic energy per molecule, one obtains values up to a thousand degrees Celsius, or so! In this region the air molecules become ionized (i.e. stripped of their electrons) by the incoming radiation, resulting in the nocturnal phenomenon of *Aurora*. This phenomenon is observed close to the two poles of the earth because the ionizing radiation becomes trapped by the earth's magnetic field, and near the poles (where the field lines enter and leave the earth's surface) the density of this radiation is highest. More generally, the layers of ionized molecules - the so-called *ionosphere* - causes disturbances to radio transmissions.

Beyond the thermosphere there is no thermopause: the atmosphere gradually merges with the vacuum of space in a region known as the *exosphere*. This has not been included in the **Fig. 1**.

We now consider what happens when short wave UV solar radiation is absorbed by the atmosphere.

2. The ozone layer

Fig. 2 shows the approximate shape of the wavelength spectrum of radiation from the sun that reaches the earth's upper atmosphere (i.e. before any is absorbed).

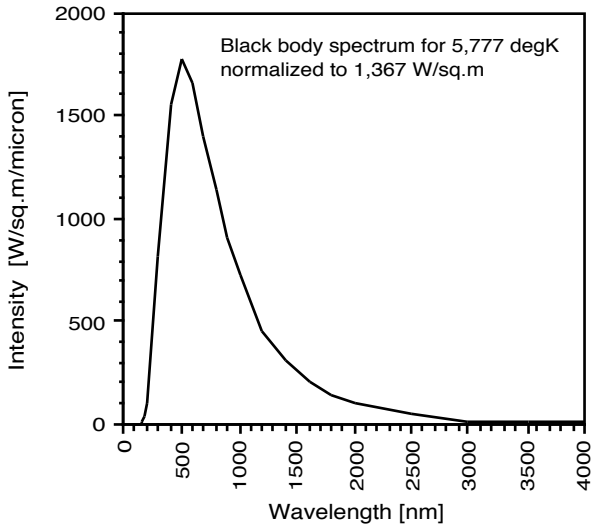


Figure 2: Approximate energy spectrum of solar radiation reaching the top of the earth's atmosphere

The part with wavelengths below about 400 nm (approximately 9% in energy) is referred to as *ultraviolet* radiation, or UV. The part in the approximate range 400 - 700 nm (approximately 55% in energy) is referred to as *visible* light, or VIS, and the part with wavelengths longer than about 700 nm (approximately 36% in energy) is referred to as *infrared* radiation, or IR. As the solar radiation descends through the earth's atmosphere energy is selectively absorbed out by different physical mechanisms among the atmospheric constituents. First, let us discuss the UV part of the spectrum.

2.1. Formation of the ozone layer

Ozone is a gas that is formed when solar ultraviolet light interacts with oxygen molecules. If the incoming solar photons have sufficiently high energy (so-called *UV-C* radiation) they tear apart the oxygen molecules, forming highly reactive oxygen *atoms*. The chemical reaction is:

$$\gamma\ (\lambda < 190\ \text{nm}) + O_2 \rightarrow O + O \qquad (11.1)$$

where γ represents a photon with wavelength less than 190 nm (this is the UV-C range of photon energies). These free oxygen atoms quickly attach themselves to neighboring oxygen molecules, forming molecules of ozone gas (O_3). The chemical reaction is:

$$O + O_2 + M \rightarrow O_3 + M \qquad (11.2)$$

In eq. (11.2), *M* represents a neighboring molecule (such as nitrogen) which must be present in order for the collision process to conserve both momentum *and* energy.

This is the way ozone is formed in the earth's atmosphere. Now, unlike the principal gases in the atmosphere – which are completely mixed together as a result of constant molecular collisions, ozone is *not* thoroughly mixed with the rest of the atmospheric constituents. Instead, it forms a *layer* in the stratosphere.

It is easy to see why this is so. The formation of ozone requires two conditions: First, a plentiful supply of UV-C photons, and second, a plentiful supply of oxygen targets. Far from the earth, there are plently of UV-C photons arriving from the sun, but there are very few oxygen molecules to stop them. Gradually, as the UV-C radiation gets closer to the earth, the density of the atmosphere increases. Therefore, more and more UV-C photons are absorbed until none are left. We thus have a situation in which at low altitudes there are plenty of oxygen molecules but no UV-C radiation to break them up (because it has all been absorbed at higher altitudes). On the other hand, at the highest altitudes there are plenty of UV-C photons but no oxygen molecules to stop them. There will accordingly be an optimal altitude where ozone is produced. This is the so-called *stratospheric ozone layer*. It is the collisions between UV-C photons and oxygen molecules that is responsible for the heating of the stratosphere, as we have seen, to temperatures that are much higher than those at the top of the troposphere.

How thick is this layer? Well, clearly, it will not have well-defined boundaries. It will start off with extremely low density at the top of the stratosphere. It will then become progressively denser as we proceed to lower altitudes, reaching a maximum and then gradually disappearing altogether. The maximum density of the ozone layer occurs at altitudes between 20 km and 30 km above sea level. Now, if we could take all of that ozone and compress it into a spherical shell at standard temperature and pressure (i.e. 273 K and 1 atmosphere), the shell would have a thickness of about 3 mm. Atmospheric physicists have defined a more convenient unit for discussing the thickness of the ozone layer. It is the *Dobson unit* (DU): 3 mm = 300 DU.

2.2 Seasonal changes in the thickness of the ozone layer

Now, in addition to being created by photons, ozone molecules can also be broken apart by photons. For this purpose the energy need not be as high as that of the UV-C photons we

discussed above. Photons in the slightly longer wavelength range 290 nm - 320 nm (so-called *UV-B* radiation) have sufficient energy. The chemical reaction here is:

$$\gamma\ (290\ \text{nm} < \lambda < 320\ \text{nm}) + O_3 \rightarrow O_2 + O \qquad (11.3)$$

The free oxygen atom that is released in reaction (11.3) quickly attaches itself to to nearby molecule, forming O_3 or N_2O. However, the important point about this reaction is that it absorbs out most (but not all!) of the harmful UV-B radiation that reaches us from the sun.

Unlike UV-C radiation, which is all absorbed out by the plentiful supply of atmospheric oxygen, some UV-B radiation does reach ground level because the ozone layer is so thin.

But the thickness of the ozone layer varies with the seasons of the year. In summer time, when the flux of destructive UV-B photons is highest, the ozone layer becomes thinner. On the other hand, in winter, the flux of UV-B is smaller and the ozone layer recovers. It reaches its maximum thickness in the springtime and its minimum thickness in the autumn. Naturally, when the ozone layer is thickest in the northern hemisphere, it is thinnest in the southern hemisphere, and *vice-versa*.

In recent years, it has been discovered that many industrially-produced gases (such as the CFCs used in air-conditioners and spray cans) can break up ozone by chemical means. These agents are believed to be responsible for the so-called *hole in the ozone layer* that has been observed in recent years in the southern hemisphere. It is this effect that has led to the world-wide ban on the use of certain kinds of refrigerant gases.

3. The emission of radiation from hot objects

In the above discussion, we saw how short wave UV-A and UV-B radiation are absorbed by oxygen and ozone, respectively. We now examine the effect on IR radiation that is brought about by the presence of water vapor and carbon dioxide in the atmosphere.

The general shape of the spectrum in **Fig. 2** is characteristic of all so-called "black" objects whether they are stars like our sun or a mere pot of boiling water on a campfire. Of course, if they are very hot indeed (like the element of a 1000 W electric heater when it is switched on) they will no longer appear black, but for this discussion it is assumed that they start off black when cold. The shape of this spectrum is caused - in a manner that is too complicated for me to explain in detail - by the acceleration and deceleration of the electrons as they vibrate back and forth in the hot object. The more energetic those vibrations will be, i.e., the hotter the black body is, the shorter will be the wavelength of the peak in **Fig. 2**. Correspondingly, the cooler the black body is, the lower will be the kinetic energy of its electrons, and the radiation they emit will peak at longer wavelengths.

There is a simple relationship between the temperature of a black body and the peak wavelength of its emission spectrum, known as *Wien's Law*. It states that *the product of the temperature (in degrees K) and the peak wavelength of the emission spectrum is a constant*. Numerically, it takes the form:

$$\lambda_{peak} \times T = 2900\ \mu\text{m K} \qquad (11.4)$$

where λ_{peak} denotes the wavelength of the peak of the emission spectrum, in units of micrometers (1 μm = 1000 nm), and T denotes the temperature of the black body in degrees Kelvin. The sun's spectrum is quite similar in shape to the emission spectrum of a black body with a temperature of approximately 6000 K. This is interpreted as the mean surface temperature of the sun because its interior is very much hotter indeed.

There is another important radiation law that we must know before we can return to our discussion of atmospheric heating. It relates the temperature of a black body to its total rate of emission of radiation. This law is known as *Stefan's Law*, and takes the simple form:

$$Q' = \sigma T^4 \qquad (11.5)$$

where Q' denotes the rate at which energy is emitted with time, expressed in Joules per per square meter per second = Watts per square meter; T denotes the temperature of the black body, in degrees Kelvin, and σ is the so-called *Stefan-Boltzmann constant* = 5.67×10^{-8} W m^{-2} K^{-4}. We are now in a position to be able to calculate the average temperature of the earth, caused by its absorption of radiation from the sun.

4. The atmospheric greenhouse effect

4.1 The "good" greenhouse effect

The sun emits radiant energy, in the form of photons, at a rate of 3.85×10^{26} J/sec. This energy spreads out uniformly in all directions, and by the time it reaches the earth (i.e. after traveling 1.5×10^{11} m) it has been diluted to an intensity of only 1360 W m^{-2}. This latter number is known as the *solar constant* for the earth, and, as we shall see, is responsible for the temperature we experience on our planet.

Let us form a first estimation of what that temperature is. In order to do this, we shall assume that the earth is a perfectly black body. That is to say, it absorbs radiation from the sun, heats up, and emits radiation to space. We can use Stefan's law to calculate what the equilibrium temperature would be in such a simplified picture.

If R is the radius of the earth and S denotes the solar constant, then the total rate at which the earth receives energy from the sun is:

$$Q'_{in} = \pi R^2 S \qquad (11.6)$$

On the other hand, the rate at which the earth emits energy to outer space is:

$$Q'_{out} = 4 \pi R^2 \sigma T^4 \qquad (11.7)$$

Equating these two rates and inserting numbers gives:

$$T = \sqrt{(S/4\sigma)} = 278 \text{ K} = 5 \text{ °C} \qquad (11.8)$$

Although this number looks extremely reasonable as an average temperature for the earth, we have made two very serious simplifications which must both be corrected. First, the earth

does not absorb radiation the way a perfectly black body does - i.e. all the radiation it receives. If it did, then photographs of the earth taken from artificial satellites would be perfectly black! In actual fact, the earth reflects approximately 30% of the sun's radiation back to space without absorbing it. This effect is called the *albedo* effect. *The albedo of a surface is the fraction of the incoming radiation that is reflected from it*. This means that the incoming energy in eq. (11.6) must be reduced by 30%. If we redo the calculation, denoting the albedo by $\alpha = 0.3$, we obtain:

$$T = \sqrt{[(1 - \alpha)\, S/4\sigma]} = 255\ K = -18\ ^{o}C \qquad (11.9)$$

This time our average temperature is clearly too low, because water could not exist in liquid form and there would be no life on earth.

The second simplification that must be corrected is our neglect of the earth's atmosphere. Although most of the atmospheric constituents are transparent to solar radiation, H_2O and CO_2 absorb *strongly* in the IR part of the spectrum. As a result, they absorb part of the radiation that is emitted by the earth, heat up and re-emit radiation - some of it back down again. The resulting equilibrium temperature is accordingly raised to +15 ^{o}C.

This heating of the earth from -18 ^{o}C to +15 ^{o}C by the presence of two trace gases in the atmosphere is known as the *greenhouse effect*, because it is similar to the way glass promotes solar heating in agricultural greenhouses. It is thanks to the greenhouse effect that the temperature of the earth is warm enough to support life. That is the good news.

4.2 The "bad greenhouse effect

The *bad news* is that about 70% of the natural greenhouse warming is caused by water vapor in the atmosphere and about 30% is due to carbon dioxide. That is to say, CO_2 is responsible for about 10 degrees of greenhouse warming. Why is that bad news? Because, the trace amount of natural CO_2 in the atmosphere is steadily increasing due to the burning of fossil fuel. This means that greenhouse-warming is increasing. Therefore, unless we can replace fossil fuel by a non-polluting alternative we may expect to start suffering serious environmental consequences – e.g. the melting of glaciers and a corresponding rise in sea level. The ideal non-polluting alternatives to fossil fuel would be solar energy for the generation of electricity, and hydrogen (generated by the electrolysis of water, using solar-generated electricity) for use as transportation fuel.

Problem set 11 (radiation)

1. A black kettle contains boiling water. What is the peak wavelength of the radiation spectrum it emits?

2. Calculate the value of the solar constant at Mars if that planet is at a mean distance of 228 million km from the sun.

3. If Mars were a perfectly black body, what would be its mean temperature?

4. The Martian atmosphere is rich in CO_2. What qualitative effect would you expect this to have on the temperature of Mars?